AN INTELLIGENT FITNESS PROGRESS MONITORING APPLICATION USING PREDICTIVE ANALYTICS

A Capstone Project

Presented to the Faculty of the

College of Computer Studies

Global Reciprocal Colleges

Caloocan City

In Partial Fulfillment

of the Requirements for the Degree

Bachelor of Science in Information Technology

Prepared by:

Calma, Kasandra

Hoquiton, John Paul

Mrs. Ann Camile M. Maupay

Adviser

November, 2026

APPROVAL SHEET

This research entitled **"An Intelligent Fitness Monitoring Application Using Predictive Analytics"** prepared and submitted by **CALMA, KASANDRA A.** and **HOQUITON, JOHN PAUL L.** in partial fulfillment of the requirements for the Subject CAPS1: Capstone Study has been examined for ORAL EXAMINATION.

ANN CAMILE M. MAUPAY, MIT
Research Adviser

PANEL OF EXAMINERS

Approved by the Committee for Oral Defense on date with a grade of _____.

NAME OF PANELIST
Chair

NAME OF PANELIST
Member

NAME OF PANELIST
Member

Accepted in partial fulfillment for the Subject CAPS2: Capstone Study

NAME
Program Head/Dean

ACKNOWLEDGMENT

First and foremost, the proponents would like to express their sincere gratitude to everyone that provided them with the necessary resources and support to complete their capstone project. The proponents would also like to thank the following people for their helpful advice and feedback throughout the process on creating this capstone project:

The GRC's Department of General Services, for allowing the proponents to conduct the study in their company.

Mr. Romark Cacho, Research and Technical Adviser, for her support, encouragement, thoughtful advice and recommendations throughout the study.

Mr. Domingo Tanael, CCS Program Head, for providing feedbacks, recommendations and his time for consultation throughout the study.

To the Members of the Panel, for the constructive comments and suggestions in improving the study.

Finally, the proponents are greatly grateful to all of the friends, classmates, and relatives for their unwavering support and encouragement throughout this journey.

DEDICATION

The proponents wholeheartedly dedicate this study to Almighty God, who has given us wisdom, strength, patience, and determination throughout the entire process of completing this project. His guidance has helped us overcome every challenge and difficulty we encountered along the way.

This study is also dedicated to our beloved families, whose unconditional love, understanding, and continuous support have served as our motivation to persevere. We are deeply grateful for their moral, emotional, spiritual, and financial support, especially during the times when we felt discouraged and exhausted.

We would also like to express our sincere appreciation to our friends, classmates, professors, and peers who shared their knowledge, encouragement and words of motivation helped us remain committed to completing this study.

Lastly, we dedicate this project to everyone who believed in our capabilities and supported us throughout this journey. This accomplishment would not have been possible without the people who continuously encouraged and inspired us to pursue our goals and bring this project to completion.

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

This chapter presents the context of our project, the objective of this study, the scope and limitation of the study, significances of the study and definition of key terms.

1.1 PROJECT CONTEXT

The growing awareness of physical fitness among Filipinos has driven a significant increase in gym memberships and structured exercise participation nationwide. As individuals pursue specific goals such as weight loss, muscle gain, and improved endurance, the demand for reliable and intelligent fitness management tools has become increasingly pronounced. Despite this growth, many gyms in the Philippines continue to operate without systematic digital tools for managing member records, tracking fitness progress, or facilitating structured trainer-member communication. This gap is particularly evident at Triple J Fitness Center, located in Valenzuela City, where current operations rely on manual and informal methods that limit service quality and member engagement.

Triple J Fitness Center serves a growing membership base in Valenzuela City, offering gym equipment and trainer-assisted workout programs for members with varying fitness goals. A preliminary consultation with the management and staff of the center revealed several persistent operational challenges: member workout logs are maintained informally, progress monitoring is inconsistent, trainer-member communication occurs without a structured system, and membership records are managed through non-digital means. These limitations have affected the center's ability to deliver consistent, data-driven fitness guidance and to retain members over the long term.

In response to these identified needs, this study proposes the development of a web-based Intelligent Fitness Progress Monitoring Application using Predictive Analytics, deployed specifically for Triple J Fitness Center. The system will serve three primary user roles: Admin, Member, and Trainer. Members will access the platform via QR code to log workouts, update fitness records, and transmit progress data to their assigned trainers. Each workout log entry is computed using standardized Metabolic Equivalent of Task (MET) values, allowing the system

to estimate calories burned per exercise based on exercise type, member body weight, and session duration. Trainers will review member logs, assess progress, and provide personalized feedback. The admin will manage user accounts, memberships, reports, and system records. By integrating predictive analytics and an AI-powered food recommendation feature which identifies food items from member-submitted photos and retrieves their corresponding nutritional values (calories, protein, carbohydrates, and fats) from the DOST Food and Nutrition Research Institute's Philippine Food Composition Table (PhilFCT), the proposed system aims to transform Triple J Fitness Center's operations into a structured, data-driven, and technology-enabled fitness environment.

1.2 OBJECTIVES OF THE STUDY

GENERAL OBJECTIVE

The general objective of this study is to design, develop, and deploy a web-based Intelligent Fitness Progress Monitoring Application using Predictive Analytics for Triple J Fitness Center in Valenzuela City by the end of Academic Year 2025-2026, integrating digital management tools for Admins, structured progress tracking for Members, and a data-driven feedback platform for Trainers.

SPECIFIC OBJECTIVE

- To develop an administrative module that enables the Admin to manage user accounts, membership records, trainer and member profiles, and generate system reports through a centralized web-based interface.
- To implement a QR code-based member access system that enables at least 90% of enrolled members to log workout sessions, record body measurements, and update fitness activity records within one month of deployment.
- To develop a workout logging module that computes estimated calories burned per exercise using standardized MET values, body weight, and session duration.

- To develop a trainer-facing module that provides trainers real-time access to member logs, progress summaries, and tools for delivering personalized feedback and workout recommendations.
- To implement a predictive analytics engine that analyzes historical member data to generate progress forecasts with a minimum accuracy threshold of 80%.
- To integrate an AI-powered food recommendation module that identifies food items from member-submitted photos and generates personalized meal suggestions using nutritional data from the DOST-FNRI's PhilFCT.

1.3 SCOPE AND LIMITATIONS OF THE STUDY

SCOPE

This study focuses on the development of the Intelligent Fitness Progress Monitoring Application, a web-based system built using Flutter (Dart) and Supabase, designed specifically for deployment at Triple J Fitness Center in Valenzuela City.

a. The system allows multiple users to access the system through three roles: Admin, Member, and Trainer.
    - The Admin manages the full scope of system operations, including user account creation and management, membership records, trainer and member data, and the generation of activity and performance reports.
    - The Member accesses the web application by scanning a QR code through which they can log workout sessions, update body measurements, record fitness activity, and transmit progress data directly to their assigned trainer.
    - The Trainer uses the platform to review member logs, monitor progress trends, and provide personalized feedback and workout recommendations.

b. The system computes estimated calories burned for each logged exercise.
    - Each logged exercise is matched against a stored MET value dataset, which the system uses together with the member's body weight and session duration to calculate the estimate.

c. The system incorporates two AI-driven features.
    - A Predictive Analytics Engine analyzes historical member data, including workout frequency, attendance records, and body measurement trends, to generate forecasts of future fitness outcomes and identify members at risk of stagnating or disengaging.
    - An AI Food Recommendation Module allows members to submit a photo of their meal, which the system analyzes to identify the food item and retrieve its corresponding calorie, protein, carbohydrate, and fat content from the DOST-FNRI PhilFCT database, then uses this data to generate personalized meal suggestions aligned with the member's weight loss, muscle gain, or maintenance objectives.

d. The system is hosted on Supabase's cloud infrastructure, ensuring real-time data synchronization across all user sessions.

LIMITATIONS

The study will not include medical diagnoses, clinical dietary prescriptions, or professional healthcare treatment recommendations; all AI-generated suggestions are informational and supplementary in nature only. The system will also not support integration with wearable fitness devices such as smartwatches or biometric trackers, as all data entry is performed manually by members through the web interface, and will not include real-time video-based workout coaching or live streaming functionality.

The other limitations of the study are that the accuracy of the AI food recommendation feature is bounded by the coverage of the PhilFCT database and the food-image identification model, as food items do not present in the database or not visually identifiable from a submitted photo fall outside the scope of automated recognition. Also, the study is limited to the design, development, and initial evaluation of the system based on the functional requirements established in consultation with Triple J Fitness Center; full organizational adoption and long-term impact assessment are beyond the scope of this study.

1.4 SIGNIFICANCE OF THE STUDY

This part of the study explained the importance of the development of the system to the following individuals:

TRIPLE J FITNESS CENTER

The primary benefactor of this study is Triple J Fitness Center. The study on implementing an Intelligent Fitness Progress Monitoring Application would primarily benefit the center's operational efficiency, member management, and its ability to offer trainers data-driven tools for delivering personalized fitness guidance, positioning it to offer a more competitive and professional service experience.

MEMBERS OF THE TRIPLE J FITNESS CENTER

The next benefactor of this study are the members of Triple J Fitness Center. The system will give them access to an organized, data-driven platform where they can track their workout routines with accurate, MET-based calorie estimates, monitor body measurements, receive AI-generated food recommendations backed by an authoritative national nutrition database, and stay informed through predictive progress insights, supporting greater accountability, consistency, and engagement in their individual fitness journeys.

TRAINERS

The other benefactor of this study are the trainers of Triple J Fitness Center. The system will provide them a centralized platform with real-time visibility into member progress, removing reliance on informal communication and enabling a more professional and evidence-based approach to client management through structured delivery of personalized recommendations and feedback.

PROPONENTS

This will serve them as the first step towards success in the field of expertise. This will also help them apply the knowledge they have gained throughout the project, contributing to the growing body of knowledge on client-based fitness information systems, AI integration in

health applications, and the application of predictive analytics within the Philippine fitness industry.

FUTURE PROPONENTS

The proposed study will serve as a reference for other or future proponents exploring similar domains, particularly future capstone and research projects on client-based fitness information systems or AI integration in health applications. It may also serve as a model for similar client-engaged research, demonstrating the practical value of integrating emerging technologies such as web-based systems, predictive analytics, and AI into student-developed capstone projects.

1.5 DEFINITION OF TERMS

Upon the implementation, the following terms that are not familiar to the users and future researchers will be the following, it has been defined clearly and conceptually for better understanding.

Admin – refers to the designated system administrator of Triple J Fitness Center, responsible for managing user accounts, membership records, trainer and member data, and system-generated reports within the proposed application.

Artificial Intelligence (AI) – refers to the application of computational methods that enable the system to analyze patterns in fitness data generate automated, context-aware outputs such as food recommendations and predictive progress forecasts.

Body Measurements – quantitative physical metrics recorded by members through the system, including body weight, height, body mass index, and circumferential measurements, used to monitor physical changes over time.

Flutter (Dart) – an open-source, cross-platform application development framework used to build the web-based frontend interface of the proposed system

Food Recommendation Module – an AI-powered feature of the proposed system that identifies food items from member-submitted photos, retrieves their corresponding nutritional values from the PhilFCT database, and generates personalized meals and dietary suggestions for members based on their declared fitness goals, including weight loss, muscle gain, or maintenance

Metabolic Equivalent of Task Value (MET) – A standardized numerical value representing the energy cost of a specific physical activity relative to resting metabolism, stored in the system's exercise dataset and used, together with member body weight and session duration, to estimate calories burned per workout log entry.

Member – an enrolled member of Triple J Fitness Center who accesses the web application through a QR code to log workout sessions, update fitness activity records, and receive AI-generated food recommendations and predictive progress insights.

Philippine Food Composition Table (PhilFCT) – the official Philippine food composition database maintained by the Department of Science and Technology Food and Nutrition Research Institute (DOST-FNRI), used by the proposed system as the reference source for the calorie, protein, carbohydrate, and fat content of identified food items.

Predictive Analytics – the use of historical fitness data, statistical methods, and machine learning algorithms to forecast future member outcomes, including projected changes in body measurements, attendance trends, and goal attainment trajectories.

QR Code Access – a quick-response code assigned to each member of Triple J Fitness Center, used as the primary method for accessing the web-based fitness management application.

Supabase – an open-source, cloud-based backend-as-a-service platform built on PostgreSQL, used to host and manage the system's database, authentication, and real-time data synchronization.

Trainer – A fitness trainer employed at Triple J Fitness Center who uses the proposed system to review member workout logs, monitor progress trends, and provide structured, data-informed feedback and workout recommendations

Workout Log – A member-generated digital record of a completed exercise session, including exercise type, sets, repetitions, duration, intensity, and MET-based estimated calories burned, stored within the system for progress monitoring and predictive analytics processing

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter shows the related literature and related studies that are significant to the proposed project. It contains facts from books, thesis, journal, article, and electronic resources and further expands the idea of the study to develop a successful project.

2.1 REVIEW OF RELATED LITERATURE

FOREIGN LITERATURE

According to Laudon and Laudon (2020). Information systems improve organizational efficiency by enabling structured data processing, storage, and automation. They also said that applying these principles to fitness management supports the development of digital platforms that replace informal, paper-based practices with organized, centralized data management.

According to Russell and Norvig (2021). Artificial intelligence is the capacity of computational systems to perform tasks that require human-like reasoning, including learning from data, recognizing patterns, and generating context-aware outputs. They also added that the integration of AI into fitness platforms enables personalized guidance, automated recommendations, and system behavior that adapts to individual user data.

According to Shatte et al. (2020). Predictive analytics techniques, including machine learning and statistical modeling, can effectively forecast future health outcomes from historical data. In the context of fitness systems, they said predictive analytics supports progress forecasting, trend detection in body composition changes, and proactive identification of members at risk of disengagement.

According to Wang et al. (2020). Mobile and web health applications significantly improve user engagement and promote more consistent health monitoring behaviors compared

to manual tracking methods. Their findings affirm the practical value of web-based platforms for delivering accessible, real-time fitness management tools to gym members.

According to Patel et al. (2022). Digital health technologies have substantially improved individuals' ability to monitor physical activity and sustain health behavior change. They also said that the effectiveness of digital platforms in bridging the gap between passive health awareness and active, consistent fitness engagement underscores the relevance of such systems for gym members.

According to Johnson et al. (2021). Individuals frequently abandon manual fitness tracking due to the time and effort involved in maintaining consistent records. They also added that this behavioral pattern reinforces the need for automated, low-friction digital systems that reduce user burden while ensuring accurate and continuous monitoring.

According to Tewari and Gupta (2020). AI-driven nutritional recommendation systems that generate personalized dietary guidance from user profile data significantly improve adherence to nutritional goals. Their findings support the integration of AI food recommendation features in fitness systems, particularly for members whose goals include weight management or muscle development.

According to Ainsworth et al. (2011). Standardized MET (Metabolic Equivalent of Task) values provide a validated basis for estimating energy expenditure from exercise type, duration, and body weight. They also said that this compendium is widely adopted in fitness and health-monitoring systems as the reference standard for calorie-burn estimation.

According to Kawano and Yanai (2014). Image-based food recognition systems, when paired with structured food composition databases, can automatically identify food items from photographs and retrieve corresponding nutritional information with high accuracy. Their

findings support the technical feasibility of combining AI-based food image recognition with an established food composition reference to automate nutritional analysis for individual users.

LOCAL LITERATURE

According to the Commission on Higher Education (2023). Digital transformation across sectors in the Philippines should be advocated, emphasizing the development of accessible, user-centered information systems that address local operational challenges. This institutional direction affirms the relevance of developing a client-based digital fitness solution as a contribution to the broader digital transformation of the Philippine fitness industry.

According to Gonzales et al. (2023). Sustained engagement with digital health tools implemented in the Philippines depends heavily on user-centered design and alignment with local user needs. Their findings underscore the importance of designing a system's interface and features in direct consultation with its end users to ensure that the system remains accessible and contextually relevant.

According to Migriño and Batangan (2021). Machine learning predictive models can be practically applied to local Philippine health datasets, confirming that predictive analytics can generate actionable forecasts from data collected within the Philippine context. This finding validates the technical viability of incorporating a predictive analytics engine into a fitness management system developed for a local fitness center.

2.2 REVIEW OF RELATED STUDIES

FOREIGN STUDIES

According to the study by Wang et al. (2020). They developed a mobile health application for tracking physical activities and found that the system significantly improved user consistency in maintaining fitness routines. However, the system lacked predictive forecasting capabilities and did not include trainer-facing features or structured feedback mechanisms, limiting its utility as a comprehensive fitness management platform.

The study of Yang et al. (2021) developed a behavior-based fitness application that improved user motivation through personalized feedback. The study confirmed the effectiveness of goal-oriented digital interventions but noted the absence of AI-driven analytical features and trainer interaction components.

The study of Shatte et al. (2020) established that predictive analytics models are capable of generating meaningful health outcome forecasts from historical datasets. While the study focused on general healthcare contexts, the methodological framework it presents is directly applicable to fitness progress forecasting.

LOCAL STUDIES

Cruz (2022) developed a web-based fitness monitoring system that improved data organization and reduced reliance on paper records. However, the system served only individual users and did not include trainer-member communication features, administrative management tools, predictive analytics, or AI-generated dietary guidance, limiting its applicability to a client-based, multi-role fitness center environment.

Reyes (2021) developed a mobile-based health tracking application that improved accessibility for fitness enthusiasts. While the platform facilitated basic activity logging, it lacked trainer interaction capabilities, role-based access, and intelligent analytical features necessary for managing a fitness center's diverse user base and operational requirements.

Santos (2020) developed a fitness tracking system that generated basic progress reports, demonstrating the value of digital data management in fitness contexts. The system did not incorporate multi-user role management, predictive analytics, or AI-based features.

2.3 SYNTHESIS OF THE REVIEWED RELATED LITERATURE AND STUDIES

The proponents believe that all references and studies mentioned in this study are similar to the current study. The proponents also believe that the same methods and techniques

used in the studies mentioned in the study can be applied to the current study. Additionally, the proponents believe that the results of the current study can be compared to the results of the references and studies mentioned in the study to provide a more comprehensive understanding of the topic.

The related literature, both foreign and local, all stated that the integration of information systems, artificial intelligence, and predictive analytics is essential to improving data management, decision support, and user engagement in health and fitness contexts (Laudon and Laudon, 2020; Russell and Norvig, 2021; Shatte et al., 2020). The references used also stated that manual and basic digital tracking systems are insufficient for sustaining consistent fitness monitoring (Johnson et al., 2021; Wang et al., 2020; Patel et al., 2022), and that AI-driven food recommendation features support members with specific dietary goals (Tewari and Gupta, 2020). Ainsworth et al. (2011) and Kawano and Yanai (2014) provide the specific technical grounding for standardized MET-based calorie estimation and image-based food recognition matched against an established food composition database.

The related studies, both foreign and local, give examples of systems that are similar to the proposed project of the proponents. In the study of Wang et al. (2020), a mobile health application improved user consistency but lacked predictive forecasting and trainer-facing features. Locally, Cruz (2022), Reyes (2021), and Santos (2020) demonstrate increasing interest in digital fitness monitoring solutions among Filipino developers; however, all three systems remain limited to single-user interfaces without multi-role management, trainer communication, or intelligent analytics capabilities. All these studies provide an insight into the benefits and importance of the development of fitness management systems with AI and predictive analytics, and the proposed project of the proponents can be developed based on the ideas presented in these studies.

The proponents can use the same technology used in these studies and develop their own system that is tailored to the specific needs of Triple J Fitness Center. In addition, the proponents can also use the findings of the studies to identify the best practices for developing their own system.

CHAPTER III

METHODOLOGY AND RESEARCH DESIGN

This chapter discusses the research design of the proposed system. It also contains, the methodology used, processes, information, diagrams that represent the proposed system.

3.1 RESEARCH METHOD AND DESIGN

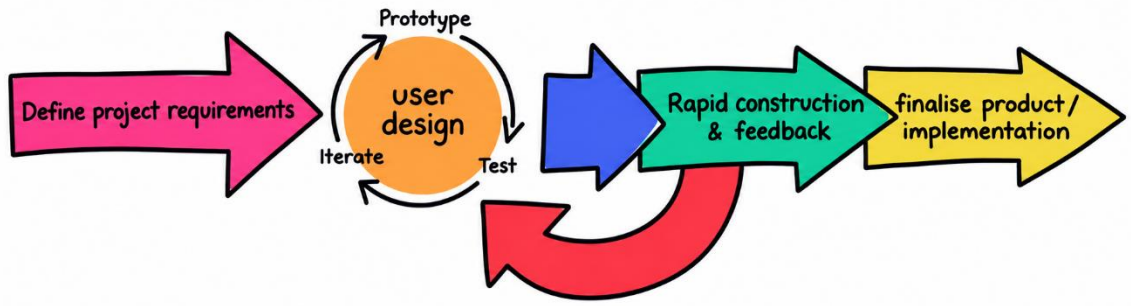


Figure 1: Research Method and Design (Rapid Application Development)

Figure 1 shows the method used in this study is the Rapid Application Development (RAD) Methodology. The RAD methodology is a software development approach that emphasizes iterative development and rapid prototyping over strict planning, allowing for continuous user involvement and progressive refinement throughout the process. The RAD methodology, as applied in this study, comprises four sequential yet iterative phases: Requirements Planning, User Design, Rapid Construction, and Cutover, with each phase producing specific outputs and using designated diagrams to document and communicate the system's structure and behavior.

The proponents began with the requirements planning phase, where the primary goal was to identify and document the operational challenges at Triple J Fitness Center and to establish the functional and non-functional requirements of the proposed system. The

researchers conducted structured interviews with the center's management, trainers, and members, conducted direct observations of existing tracking and administrative practices, and reviewed available fitness records and membership documentation. This phase directly addressed the problems identified by the study and served as the foundation for the system's three-role design involving Admin, Member, and Trainer.

3.2 CONCEPTUAL FRAMEWORK

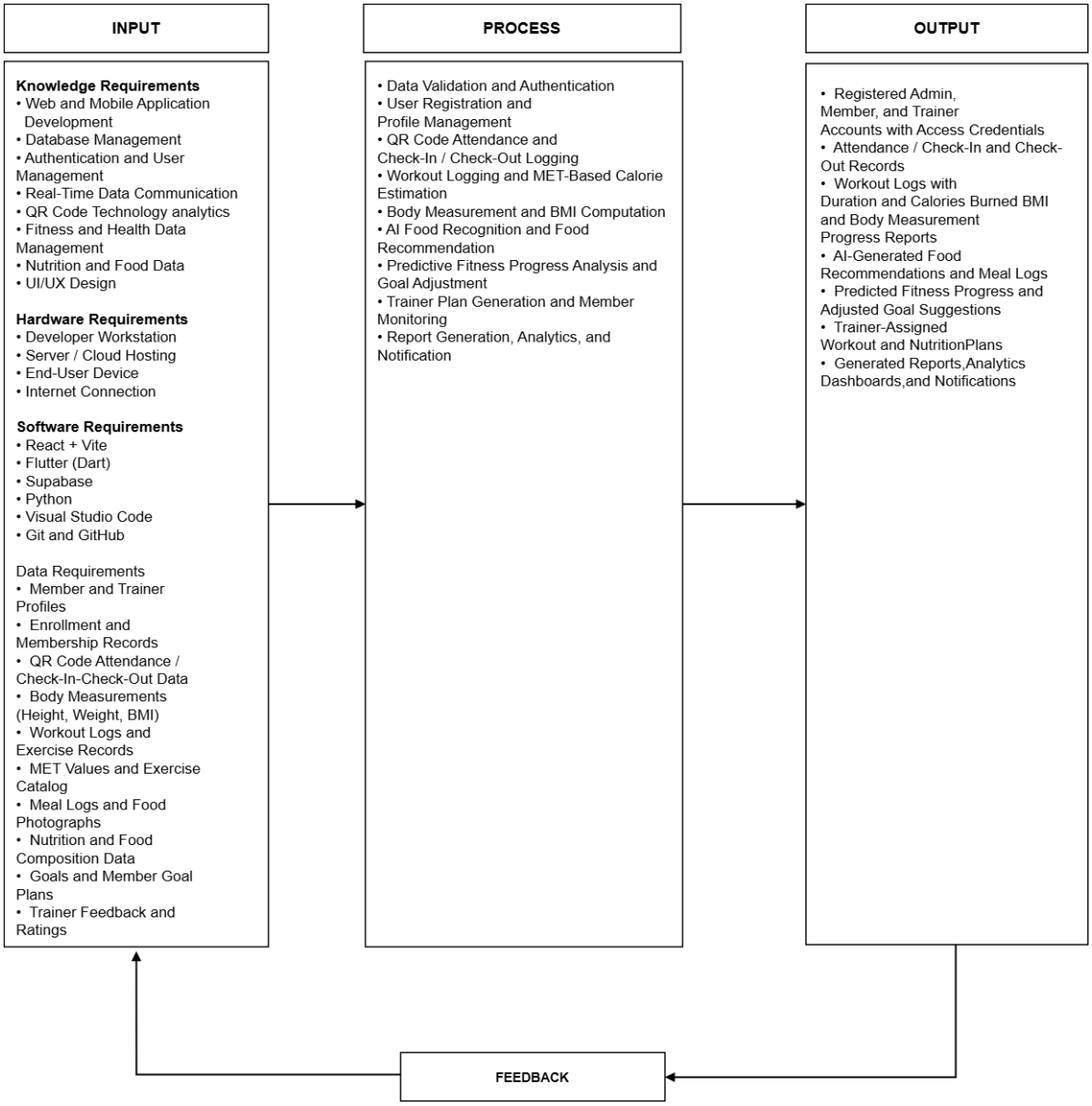


Figure 2: Input Process Output (IPO)

The Intelligent Fitness Progress Monitoring Application Using Predictive Analytics follows the Input-Process-Output (IPO) model. For the Input, the project integrates the Knowledge Requirements, which include web and mobile application development, database management, authentication and user management, real-time data communication, QR code technology analytics, fitness and health data management, nutrition and food data, and UI/UX design. It also includes the necessary Hardware Requirements, such as the developer workstation, server or cloud hosting, end-user devices, and internet connection. The Software Requirements include React with Vite, Flutter using Dart, Supabase, Python, Visual Studio Code, and Git and GitHub, which provide the necessary tools and technologies for developing and managing the system. In addition, the Data Requirements consist of the actual information that the system collects and processes, including member and trainer profiles, enrollment and membership records, QR code attendance and check-in and check-out data, body measurements such as height, weight, and BMI, workout logs and exercise records, MET values and the exercise catalog, meal logs and food photographs, nutrition and food composition data, member goals and goal plans, and trainer feedback and ratings.

The Process covers the operational functions performed by the system. It begins with data validation and authentication, which verifies user identities and enforces role-based access for Admin, Member, and Trainer accounts. This is followed by user registration and profile management, QR code attendance and check-in and check-out logging, and workout logging with MET-based calorie estimation. The system also performs body measurement and BMI computation, AI-powered food recognition and food recommendation using the nutrition and food composition database, and predictive fitness progress analysis with goal adjustment based on recorded trends. Trainer plan generation with member monitoring, together with report generation, analytics, and notification, complete the process. These processes are carried out iteratively following the Rapid Application Development (RAD) methodology presented in Figure 1, allowing continuous refinement based on user feedback and system performance.

The Output phase produces the Intelligent Fitness Progress Monitoring Application Using Predictive Analytics, which provides a structured platform for managing fitness-related information and monitoring member progress. The system generates registered Admin, Member, and Trainer accounts with access credentials; attendance and check-in and check-out

records; workout logs with duration and calories burned; BMI and body measurement progress reports; AI-generated food recommendations and meal logs; predicted fitness progress with adjusted goal suggestions; trainer-assigned workout and nutrition plans; and generated reports, analytics dashboards, and notifications. These outputs are intended to support more organized fitness management, data-driven progress monitoring, and improved communication between members and trainers.

The Feedback component provides a continuous mechanism for improving the system. As shown in Figure 2, feedback is drawn from the Output stage and returned to the Input and Process stages, so that user evaluation of the developed system can be used to identify areas for improvement and guide necessary adjustments to the requirements, design, development, and testing processes. This allows the system to be refined based on actual user needs and system performance.

3.3 HARDWARE AND SOFTWARE REQUIREMENTS

Table 1: Hardware Requirements

Component	Minimum Requirements	Recommended Requirements
Processor (CPU)	Intel Core i5 or AMD Ryzen 5	Intel Core i7 or AMD Ryzen 7
RAM	8 GB	16 GB
Storage (Primary)	256 GB SSD	512 GB SSD
Backup Storage	500 GB External HDD or Cloud Storage	1TB External Storage or Automated Cloud Backup
Display	19” LED Monitor	24” IPS Monitor
Internet Connection	10 Mbps Stable Connection	50+ Mbps Fiber Connection

Camera	Functional built-in or external camera	1080p HD Camera
Mobile Device	Android 7.0, quad-core 1.4 GHz, 2 GB RAM, 200MB, 5.0" 720×1280 display	Android 10+, octa-core 1.8 GHz+, 4GB RAM, 500MB, 6.0" 1080×2340 display, HD camera
Wired Networking	Ethernet LAN (RJ45) & Router	

Table 1 shows the hardware requirements needed to support the development and operation of the Intelligent Fitness Progress Monitoring Application. The minimum requirements provide the necessary specifications for basic development, testing, and system access, while the recommended requirements provide better performance and smoother operation. Components such as the processor and RAM support application development, data processing, and predictive analytics tasks, while SSD storage provides faster system loading and data access. Backup storage helps protect important project and system data, while a stable internet connection ensures reliable access to the cloud-based application and real-time synchronization. Wired and wireless networking allow the mobile application to communicate with the cloud database and the AI service during attendance logging and food image submission. A server or cloud hosting instance is required to run the database and the AI service that performs food recognition and predictive analysis. The camera is necessary for QR code scanning and for capturing food photos used by the AI food recommendation feature. End-user smartphones or tablets allow Members and Trainers to access the mobile application, while the Admin accesses the web-based dashboard through a computer with a display monitor. Overall, these hardware requirements provide the necessary resources for the reliable development, deployment, and use of the proposed system.

Table 2: Software Requirements

Software	**Purpose**
React + Vite	Used to develop the web-based Admin interface and provide a fast development environment for the administrative dashboard
Tailwind CSS and Radix UI	Used to build the responsive user interface and reusable components of the Admin dashboard
Recharts	Used to generate the charts and analytics visualizations displayed in the Admin dashboard
Node.js and Express	Used as the runtime environment for the Admin application and as the server that handles notification and broadcast requests
Flutter (Dart)	Used to develop the Member and Trainer applications and provide a responsive cross-platform user interface
Mobile scanner, Camera, and Image picker	Used to implement QR code scanning for attendance and to capture food photographs for AI-based recognition
Riverpod and go_router	Used for state management and navigation within the Member and Trainer applications
Supabase	Used as the backend platform for database management, authentication, data storage, and real-time data synchronization
Python	Used for data processing, predictive analytics, machine learning, and AI-related system functions

Google Gemini API (google-generativeai)	Used as the AI engine for image-based food recognition and for generating food and goal recommendations
Visual Studio Code	Used by the proponents for writing, editing, debugging, and managing the application's source code
Operating System	Windows 10 or 11 (64-bit) for development; Android 8.0 or iOS 12 and higher for end-user mobile devices
scikit-learn	Used to train and run the machine learning models for predictive fitness progress analysis and retention risk assessment
Git and GitHub	Used for version control, source code management, collaboration, and tracking changes throughout the development process

Table 2 shows the software requirements needed for the development and operation of the proposed Intelligent Fitness Progress Monitoring Application. React + Vite is used to develop the web-based Admin interface, supported by Tailwind CSS and Radix UI for its user interface components and Recharts for its charts and analytics visualizations, while Node.js and Express serve as the runtime and backend server for the Admin application. Flutter (Dart) is used to develop the Member and Trainer applications, with Riverpod and go_router providing state management and navigation, and Mobile scanner, camera, and Image picker enabling QR code scanning and the capture of food photographs. Supabase serves as the backend platform for database management, authentication, data storage, row-level security, and real-time data synchronization. Python is used for data processing, machine learning, and AI-related functions of the system, scikit-learn is used to train and run the machine learning models for predictive fitness progress analysis and retention risk assessment, while the Google Gemini API is used for image-based food recognition and for generating food and goal recommendations. Visual Studio Code is used for writing, editing, debugging, and managing the source code, while Git and GitHub support version control, collaboration, and tracking of changes throughout the development process. Overall, these software requirements provide the necessary tools and technologies for developing, managing, and operating the proposed system.

3.4 CONTEXT DIAGRAM

EXISTING

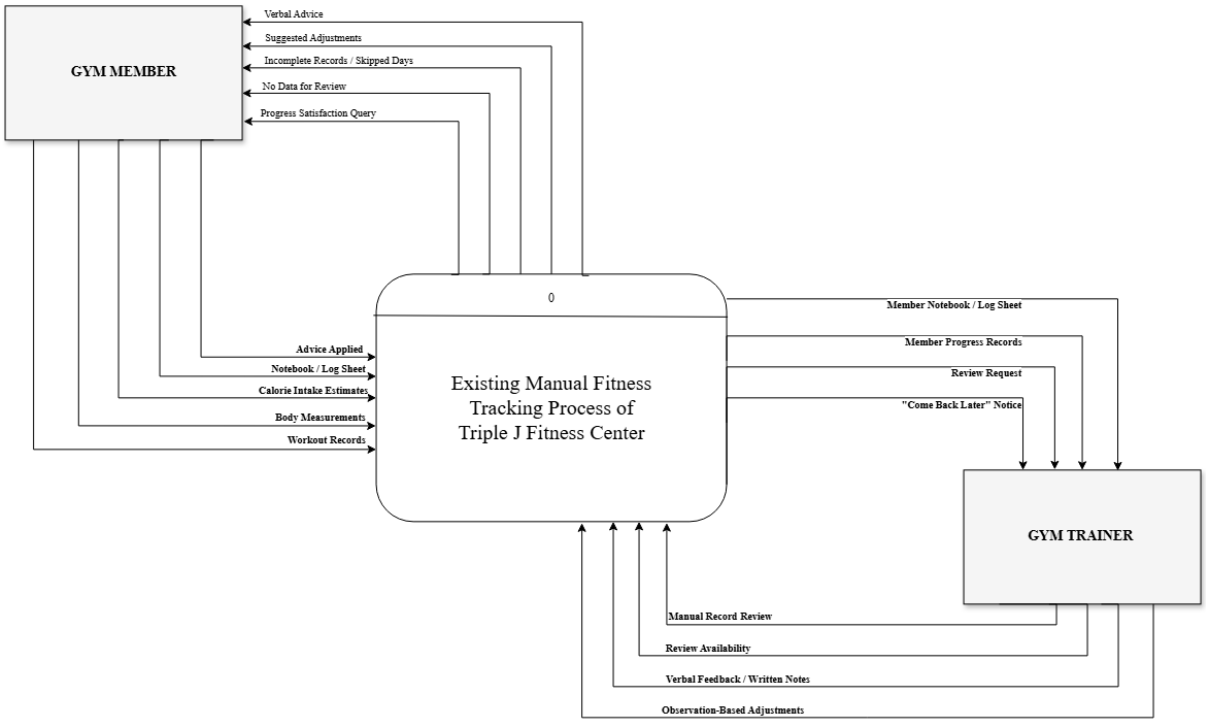


Figure 3: Context Diagram of Existing System

Figure 3 presents the context diagram of the existing system, which shows the manual fitness tracking process currently being practiced at Triple J Fitness Center. The entire manual process is represented by a single process, labeled 0, and it exchanges data with two external entities: the Gym Member and the Gym Trainer or Staff. Unlike the proposed system, the existing process does not involve any automated database, software application, or third-party service, since all records are kept manually in notebooks, spreadsheets, and log sheets.

The Gym Member provides the system with workout records containing the exercise name, sets, and repetitions, as well as body weight and body measurements obtained manually using a weighing scale and tape measure. The member also attempts to calculate calorie intake using online tools or estimates, maintains a notebook or log sheet, and later reports whether the

verbal advice given was actually applied. In return, the existing system gives the member verbal advice from the trainer, informal suggested adjustments based on the trainer's observation, a notice that the records are incomplete or that days were skipped, a notice that no data is available for review, and a question regarding whether the member is satisfied with the progress achieved. Because the process depends entirely on manual recording and personal observation, the member is unable to verify whether the progress being made is accurate, which often results in the member abandoning tracking altogether.

The Gym Trainer or Staff provides the system with a manual review of the member's records, an indication of whether the trainer has time to perform the review, verbal feedback or written notes, and adjustments based on personal observation rather than measured data. In return, the existing system provides the trainer with the member's notebook or log sheet, the member's progress records which are frequently incomplete, the request for review, and a notice that the member should come back later when the trainer is unavailable.

Overall, the context diagram of the existing system shows that the current practice relies entirely on manual records and informal communication between the member and the trainer. There is no formal monitoring, no automated computation, and no data-driven basis for the advice provided, which are the limitations that the proposed system intends to address.

PROPOSED

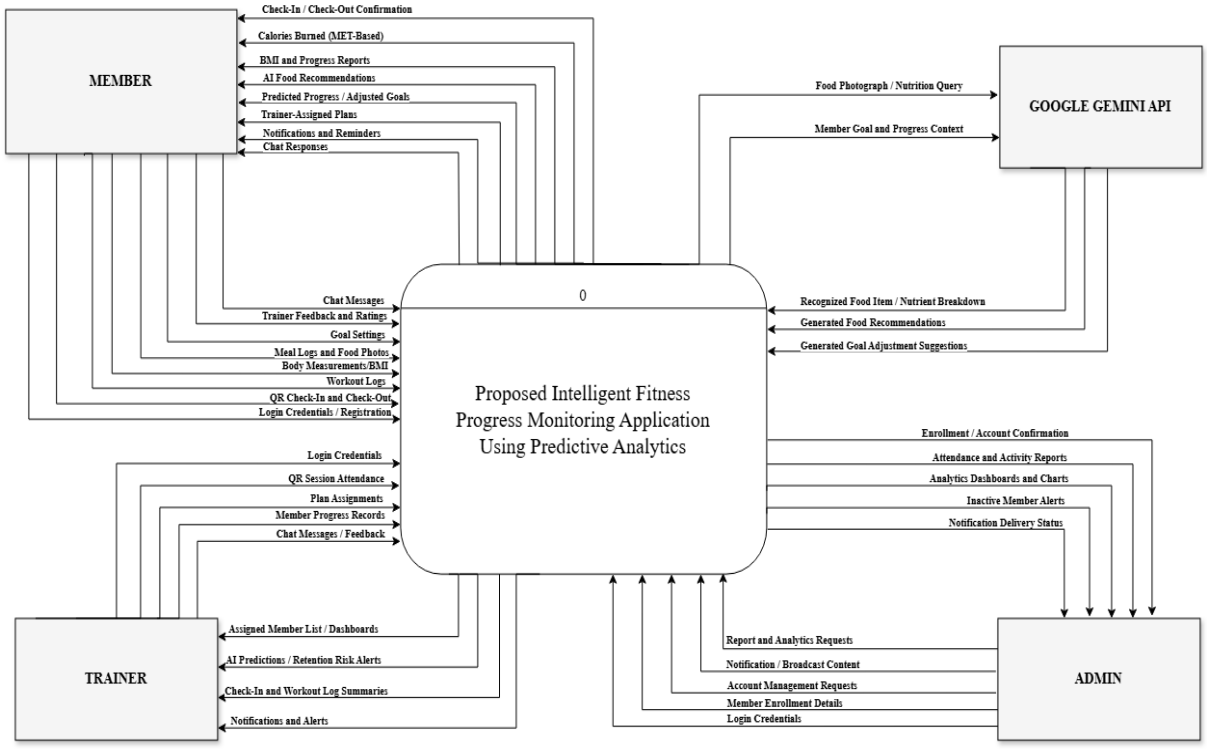


Figure 4: Context Diagram of Proposed System

Figure 4 presents the context diagram of the proposed Intelligent Fitness Progress Monitoring Application Using Predictive Analytics. The entire proposed system is represented by a single process, labeled 0, and it exchanges data with four external entities: the Member, the Trainer, the Admin, and the Google Gemini API.

The Member provides the system with login credentials and registration details, QR code check-in and check-out data, workout logs containing the exercise performed, sets, repetitions, and duration, body measurements such as height and weight, meal logs together with food photographs, goal settings, trainer feedback and ratings, and chat messages addressed to the assigned trainer. In return, the system provides the member with check-in and check-out confirmations, the computed calories burned using MET-based estimation, BMI and progress reports presented through charts, AI-generated food recommendations, predicted fitness progress with adjusted goal suggestions, trainer-assigned workout and nutrition plans, notifications and reminders, and chat responses from the trainer.

The Trainer provides the system with login credentials, QR code session attendance, workout and nutrition plan assignments for assigned members, member progress records, and chat messages and feedback. In return, the system provides the trainer with the list of assigned members and their progress dashboards, AI-generated progress predictions and retention risk alerts, summaries of member check-ins and workout logs, chat messages from members, and notifications and alerts.

The Admin provides the system with login credentials, member enrollment details, account management requests such as creating accounts, assigning trainers, and deleting users, notification and broadcast content, and report and analytics requests. In return, the system provides the admin with enrollment and account confirmations, attendance and activity reports, analytics dashboards and charts, inactive member alerts, and the delivery status of notifications.

The Google Gemini API, acting as an external third-party service, receives food photographs and nutrition queries from the system and returns the recognized food item with its nutrient breakdown and generated food recommendations. It likewise receives the member's goal and progress context and returns generated goal adjustment suggestions. The system's predictive analytics component, which uses scikit-learn, and the Supabase platform, which serves as the system's database and authentication backend, are considered part of the system rather than external entities.

Overall, the context diagram of the proposed system shows that the manual recording and informal communication observed in the existing system are replaced by an automated platform that captures data through QR codes and mobile inputs, processes them through predefined and AI-based computations, stores them in a centralized cloud database, and returns accurate, data-driven outputs to the member, the trainer, and the admin.

3.5 DATA FLOW DIAGRAM

EXISTING

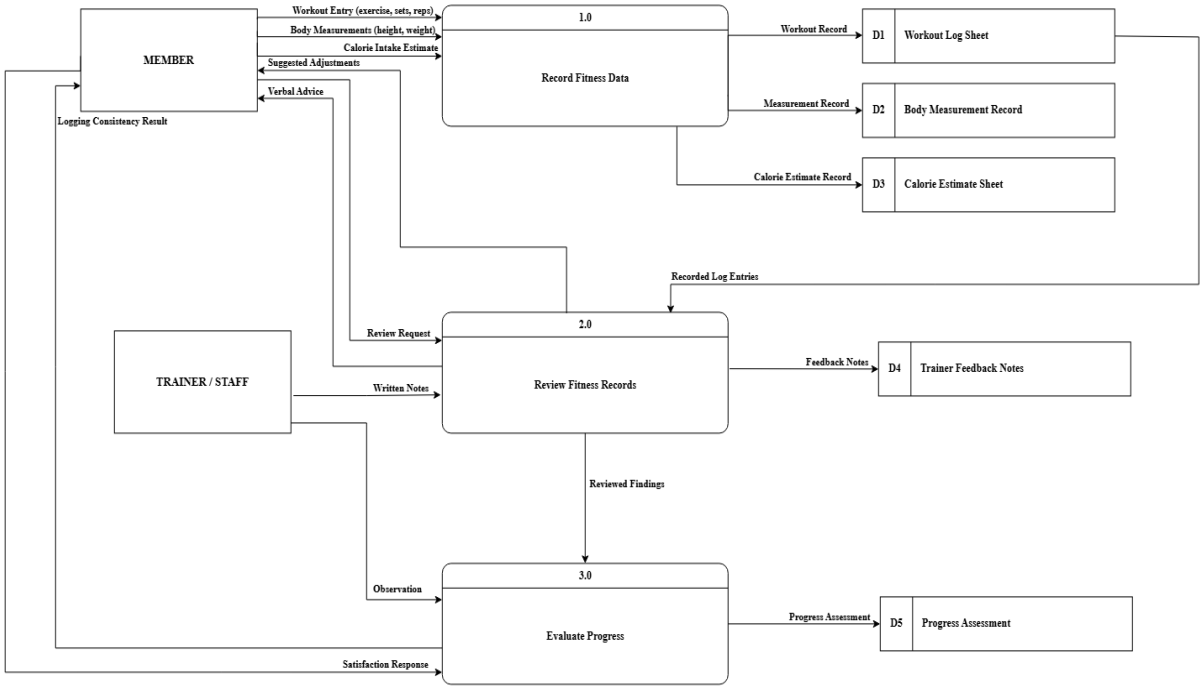


Figure 5: Data Flow Diagram of Existing System

Figure 5 presents the data flow diagram of the existing system of Triple J Fitness Center. The existing system is composed of three processes, namely 1.0 Record Fitness Data, 2.0 Review Fitness Records, and 3.0 Evaluate Progress, and it exchanges data with two external entities: the Member and the Gym Trainer or Staff. All data in the existing system are stored manually in the Workout Log Sheet, the Body Measurement Record, the Calorie Estimate Sheet, the Trainer Feedback Notes, and the Progress Assessment.

In the first process, the Member supplies the system with a workout entry containing the exercise name, the number of sets, and the number of repetitions, together with body measurements obtained manually using a weighing scale and tape measure, and a calorie intake estimate taken from online tools or personal estimation. The system records these into the Workout Log Sheet, the Body Measurement Record, and the Calorie Estimate Sheet.

In the second process, the recorded log entries are retrieved and submitted to the trainer together with the review request of the member. The Gym Trainer or Staff provides written notes, and the system produces feedback notes and returns verbal advice and suggested adjustments to the member.

In the third process, the reviewed findings are evaluated against the member's weekly logging consistency and satisfaction response. The system produces a progress assessment and returns the logging consistency result to the member. Because the entire process depends on manual recording and personal observation, the resulting assessment has no data-driven basis, which is the limitation that the proposed system addresses.

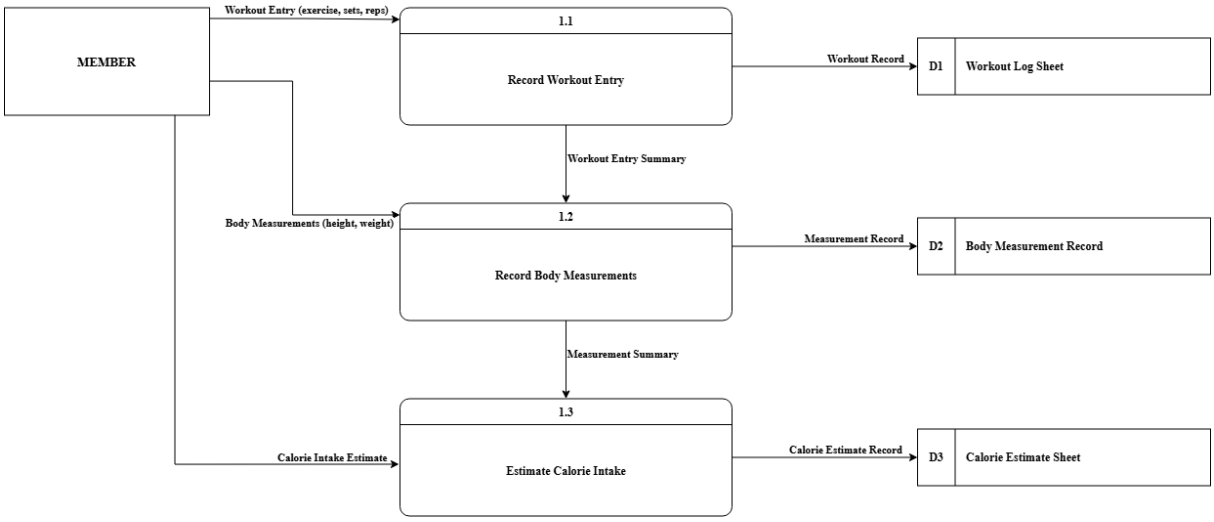


Figure 6: Data Flow Diagram of Process 1 (Existing)

Figure 6 represents the data flow diagram of process 1 of the existing system of Triple J Fitness Center. This diagram decomposes the Record Fitness Data process into three sub-processes: 1.1 Record Workout Entry, 1.2 Record Body Measurements, and 1.3 Estimate Calorie Intake.

In sub-process 1.1, the Member provides a workout entry containing the exercise name, the number of sets, and the number of repetitions, which is written into the Workout Log Sheet.

The resulting workout entry summary is passed to sub-process 1.2, where the Member records body measurements such as weight and height taken manually with a weighing scale and tape measure. These are written into the Body Measurement Record and the resulting measurement summary is passed to sub-process 1.3.

In sub-process 1.3, the Member attempts to calculate calorie intake using online tools or personal estimation, and the resulting value is written into the Calorie Estimate Sheet. Because all three sub-processes are performed by hand, the records they produce are prone to missing days and inaccurate values, which frequently leaves no data available for review by the trainer.

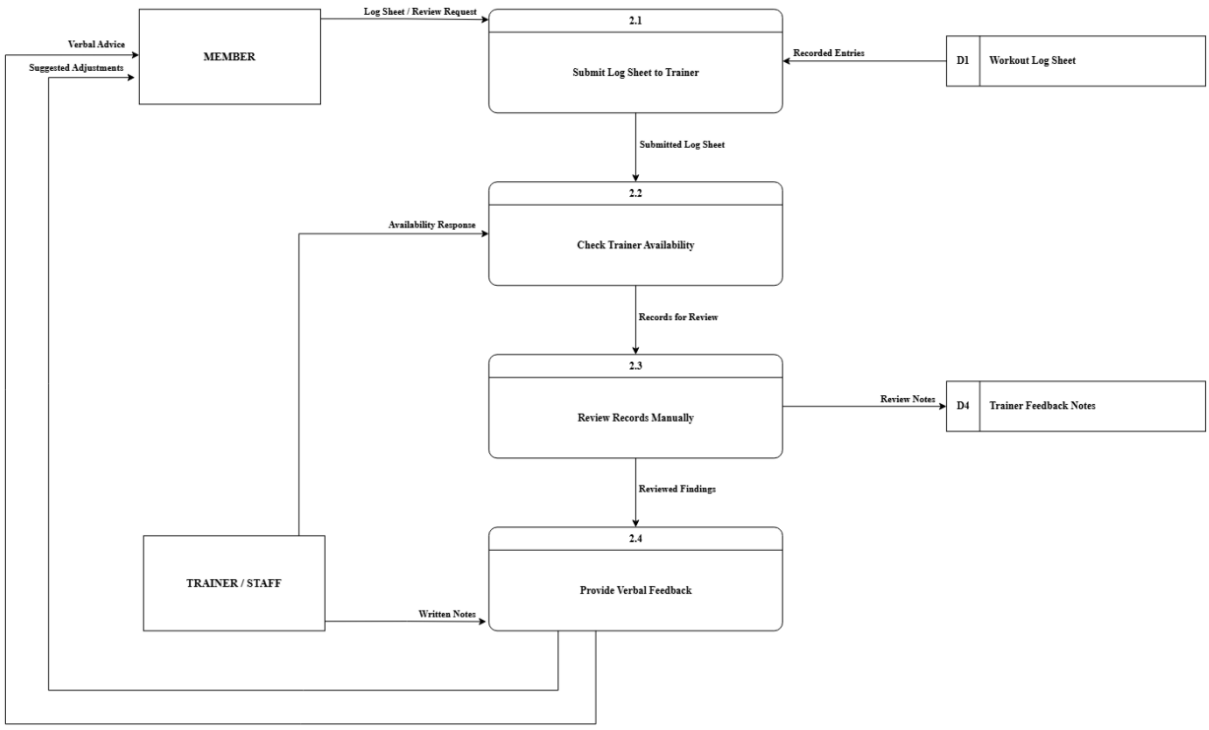


Figure 7: Data Flow Diagram of Process 2 (Existing)

Figure 7 represents the data flow diagram of process 2 of the existing system of Triple J Fitness Center. This diagram decomposes the Review Fitness Records process into four sub-

processes: 2.1 Submit Log Sheet to Trainer, 2.2 Check Trainer Availability, 2.3 Review Records Manually, and 2.4 Provide Verbal Feedback.

In sub-process 2.1, the Member submits the log sheet together with a review request, while the recorded entries are retrieved from the Workout Log Sheet. The submitted log sheet is passed to sub-process 2.2, where the system determines whether the Gym Trainer or Staff has time to perform the review, based on the availability response of the trainer. If the trainer is unavailable, the member waits or comes back later.

Once the trainer is available, the records for review are passed to sub-process 2.3, where the Gym Trainer or Staff reviews the records manually and the resulting review notes are written into the Trainer Feedback Notes. The reviewed findings are then passed to sub-process 2.4, where the trainer's written notes are converted into verbal advice and suggested adjustments, which are returned to the Member.

Because the review depends entirely on the trainer's availability and personal judgement, the advice produced is not data-driven, and the member cannot verify whether the progress being made is accurate.

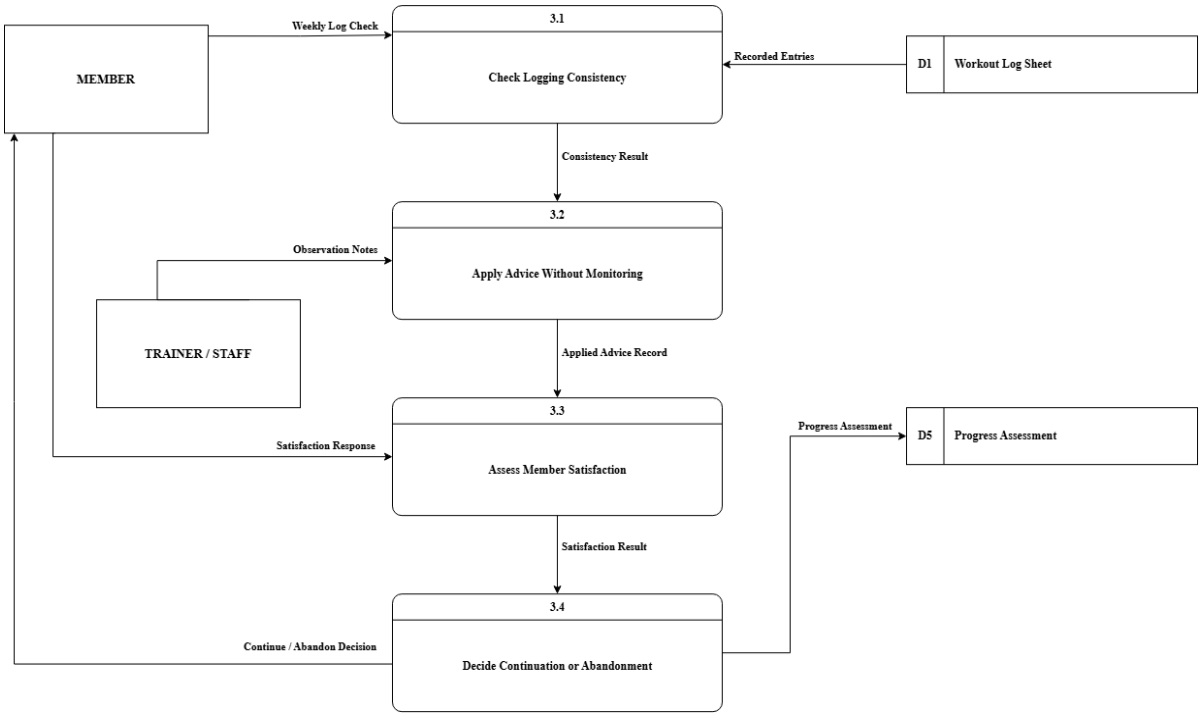


Figure 8: Data Flow Diagram of Process 3 (Existing)

Figure 8 represents the data flow diagram of process 3 of the existing system of Triple J Fitness Center. This diagram decomposes the Evaluate Progress process into four sub-processes: 3.1 Check Logging Consistency, 3.2 Apply Advice Without Monitoring, 3.3 Assess Member Satisfaction, and 3.4 Decide Continuation or Abandonment.

In sub-process 3.1, the Member's weekly log check is compared against the recorded entrie retrieved from the Workout Log Sheet, producing a consistency result. This result is passed to sub-process 3.2, where the Member applies the advice previously received from the trainer without any formal monitoring, supported only by the observation notes of the Gym Trainer or Staff. The applied advice record is passed to sub-process 3.3.

In sub-process 3.3, the Member provides a satisfaction response, which is evaluated against the applied advice record to produce a satisfaction result. This result is passed to sub-process 3.4, where the decision to continue or abandon tracking is made and the progress assessment is written into the Progress Assessment.

Because the evaluation has no measurable baseline, the decision is based on the member's feeling rather than on recorded data. When the member is not satisfied, the outcome of the existing process is that the member abandons tracking altogether.

PROPOSED

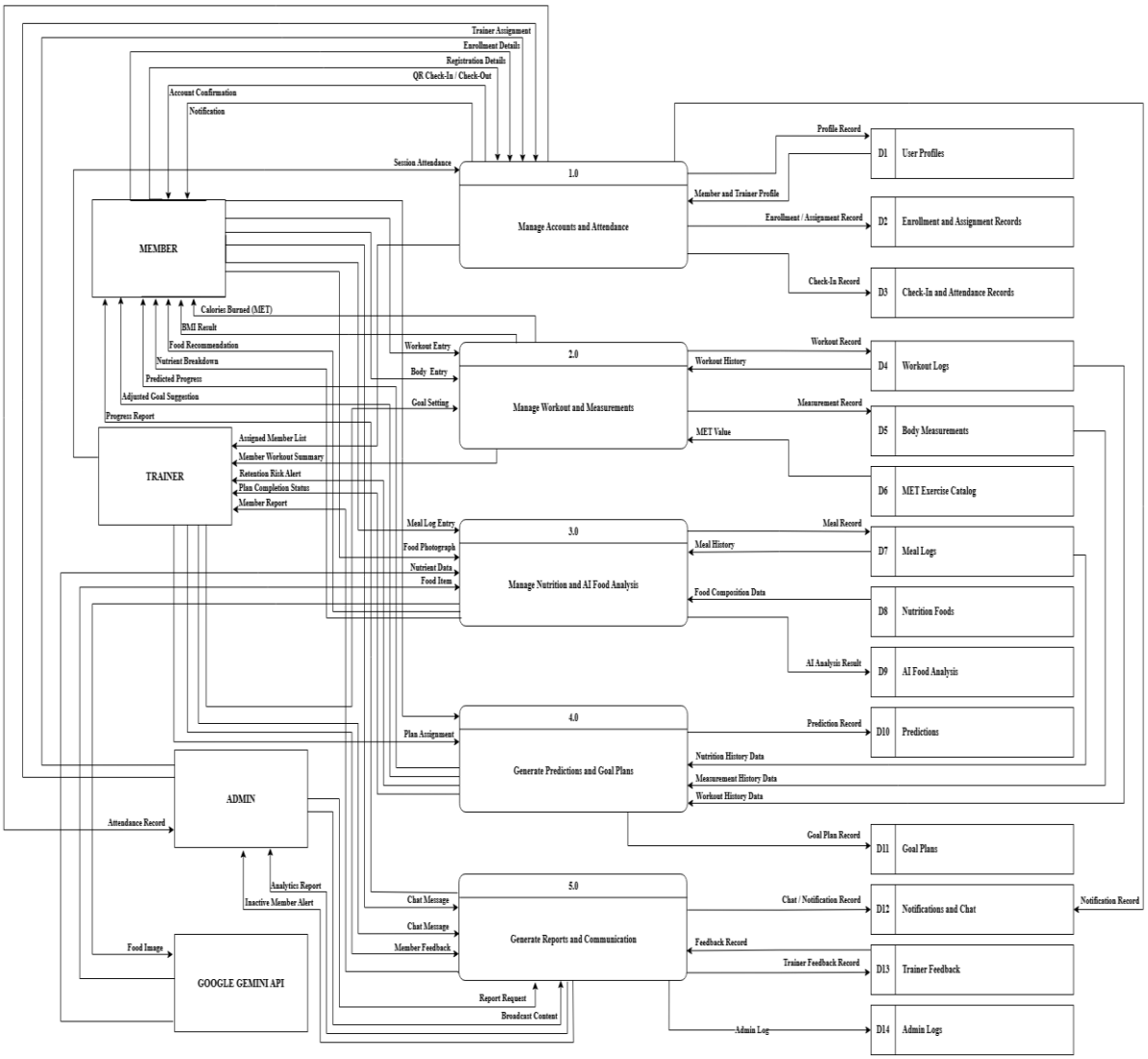


Figure 9: Data Flow Diagram of Proposed System

Figure 9 presents the data flow diagram of the proposed Intelligent Fitness Progress Monitoring Application Using Predictive Analytics. The proposed system is composed of five

processes, namely 1.0 Manage Accounts and Attendance, 2.0 Manage Workout and Measurements, 3.0 Manage Nutrition and AI Food Analysis, 4.0 Generate Predictions and Goal Plans, and 5.0 Generate Reports and Communication. The system exchanges data with four external entities: the Member, the Trainer, the Admin, and the Google Gemini API.

In process 1.0, the Admin provides enrollment details and trainer assignment requests, while the Member provides registration details and QR code check-in and check-out data. The process stores the profile records in the User Profiles, the enrollment and assignment records in the Enrollment and Assignment Records, the check-in record in the Check-In and Attendance Records, and returns the account confirmation, attendance record, assigned member list, and notification to the users.

In process 2.0, the Member provides a workout entry and a body measurement entry. The process retrieves the MET value from the MET Exercise Catalog, computes the calories burned, and stores the workout record in the Workout Logs and the measurement record in the Body Measurements, while returning the calories burned and the BMI result to the member and the member workout summary to the trainer.

In process 3.0, the Member provides a meal log entry and a food photograph. The food image and nutrition query are sent to the Google Gemini API, which returns the recognized food item and the nutrient breakdown data. The process stores the meal record in the Meal Logs and the analysis result in the AI Food Analysis, and returns the food recommendation and the nutrient breakdown to the member.

In process 4.0, the workout, measurement, and nutrition histories are analysed to produce the prediction record and the goal plan record. The process returns the predicted fitness progress and the adjusted goal suggestion to the member, and the retention risk alert and the plan completion status to the trainer.

In process 5.0, the chat messages and trainer ratings of the member, the feedback of the trainer, and the report request and broadcast content of the admin are processed to produce the progress report, the member report, the analytics report, and the inactive member alert.

Overall, the data flow diagram of the proposed system shows that the manual recording and informal communication of the existing system are replaced by an automated platform that captures data through QR codes and mobile inputs, processes it through predefined and AI-based computations, stores it in a centralized cloud database, and returns accurate, data-driven outputs to the member, the trainer, and the admin.

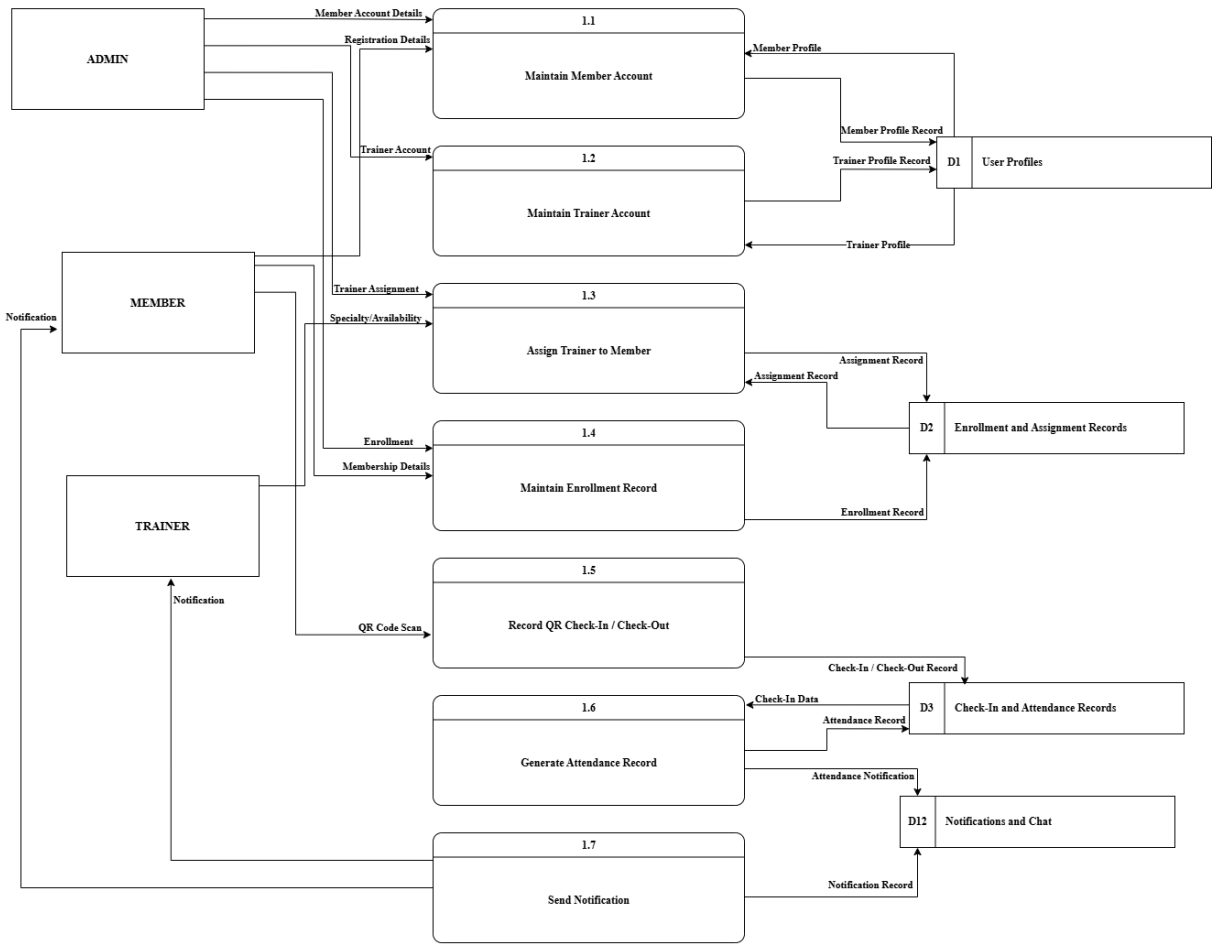


Figure 10: Data Flow Diagram of Process 1 (Proposed)

Figure 10 represents the data flow diagram of process 1 of the proposed system. This diagram decomposes the Manage Accounts and Attendance process into seven sub-processes:1.1 Maintain Member Account, 1.2 Maintain Trainer Account, 1.3 Assign Trainer to Member,1.4 Maintain Enrollment Record, 1.5 Record QR Check-In / Check-Out, 1.6 Generate Attendance Record, and 1.7 Send Notification.

In sub-process 1.1, the Admin provides the member account details while the Member provides the registration details. The process writes the member profile record into the User Profiles and returns the member profile for verification. In sub-process 1.2, the Admin provides the trainer account details, which are written into the User Profiles.

In sub-process 1.3, the Admin provides the trainer assignment while the Trainer provides the specialty and availability. The assignment record is written into the Enrollment and Assignment Records. In sub-process 1.4, the Admin provides the enrollment details and the Member provides the membership details, which are written into the same data store.

In sub-process 1.5, the Member provides the QR code scan, which is written into the Check-In and Attendance Records as a check-in and check-out record. In sub-process 1.6, the check-in data are retrieved and the attendance record is generated and stored, together with the attendance notification. In sub-process 1.7, the notification record is written into the Notifications and Chat and returned to the Member and the Trainer.

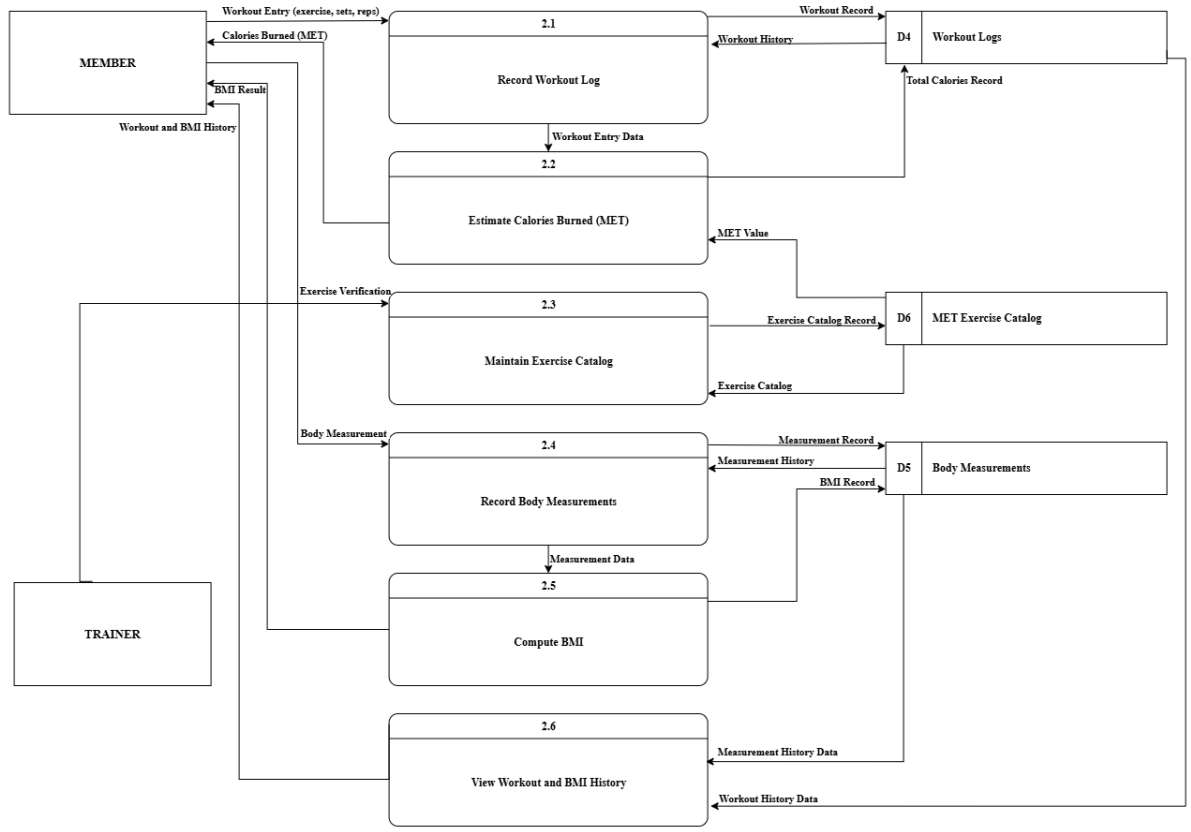


Figure 11: Data Flow Diagram of Process 2 (Proposed)

Figure 11 represents the data flow diagram of process 2 of the proposed system. This diagram decomposes the Manage Workout and Measurements process into six sub-processes: 2.1 Record Workout Log, 2.2 Estimate Calories Burned (MET), 2.3 Maintain Exercise Catalog, 2.4 Record Body Measurements, 2.5 Compute BMI, and 2.6 View Workout and BMI History.

In sub-process 2.1, the Member provides a workout entry containing the exercise name, the number of sets, the number of repetitions, and the duration, which is written into the Workout Logs. The workout entry data are passed to sub-process 2.2, where the MET value is retrieved from the MET Exercise Catalog to compute the calories burned. The computed value is returned to the Member and stored in the Workout Logs as the total calories record.

In sub-process 2.3, the Trainer provides the exercise verification and the exercise catalog record is maintained in the MET Exercise Catalog. In sub-process 2.4, the Member provides the body measurement entry, which is written into the Body Measurements together with the measured date.

The measurement data are passed to sub-process 2.5, where the BMI is computed from the recorded height and weight, returned to the Member, and stored in the Body Measurements. In sub-process 2.6, the workout history data and the measurement history data are retrieved and returned to the Member as the workout and BMI history.

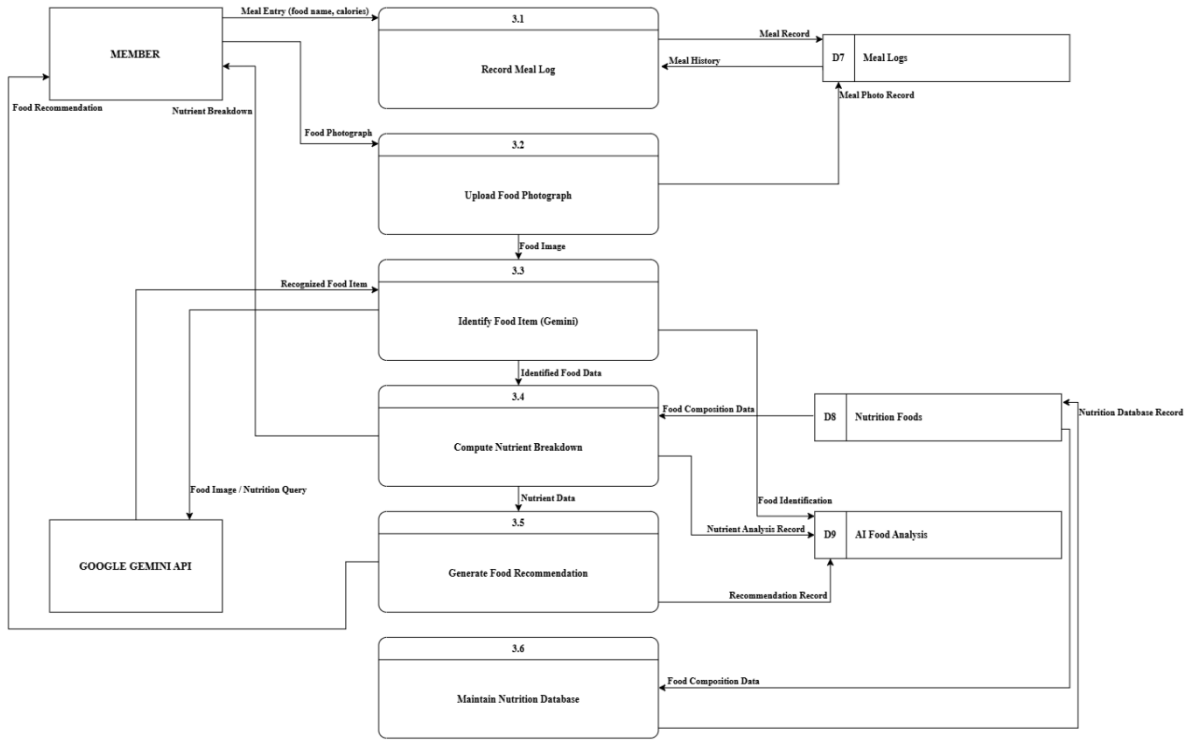


Figure 12: Data Flow Diagram of Process 3 (Proposed)

Figure 12 represents the data flow diagram of process 3 of the proposed system. This diagram decomposes the Manage Nutrition and AI Food Analysis process into six sub-processes: 3.1 Record Meal Log, 3.2 Upload Food Photograph, 3.3 Identify Food Item (Gemini), 3.4 Compute Nutrient Breakdown, 3.5 Generate Food Recommendation, and 3.6 Maintain Nutrition Database.

In sub-process 3.1, the Member provides a meal entry containing the food name, the meal type, and the calories, which is written into the Meal Logs. In sub-process 3.2, the Member provides the food photograph, which is stored as the meal photo record, and the food image is passed to sub-process 3.3.

In sub-process 3.3, the food image and the nutrition query are sent to the Google Gemini API, which returns the recognized food item. The result is written into the AI Food Analysis and the identified food data are passed to sub-process 3.4.

In sub-process 3.4, the food composition data are retrieved from the Nutrition Foods to compute the nutrient breakdown, which is returned to the Member and written into the AI Food Analysis. In sub-process 3.5, the nutrient data are used to generate the food recommendation, which is returned to the Member and stored in the AI Food Analysis.

In sub-process 3.6, the food composition data retrieved from the Nutrition Foods are used to maintain the nutrition database record, ensuring that the food reference data used by the AI analysis remain accurate.

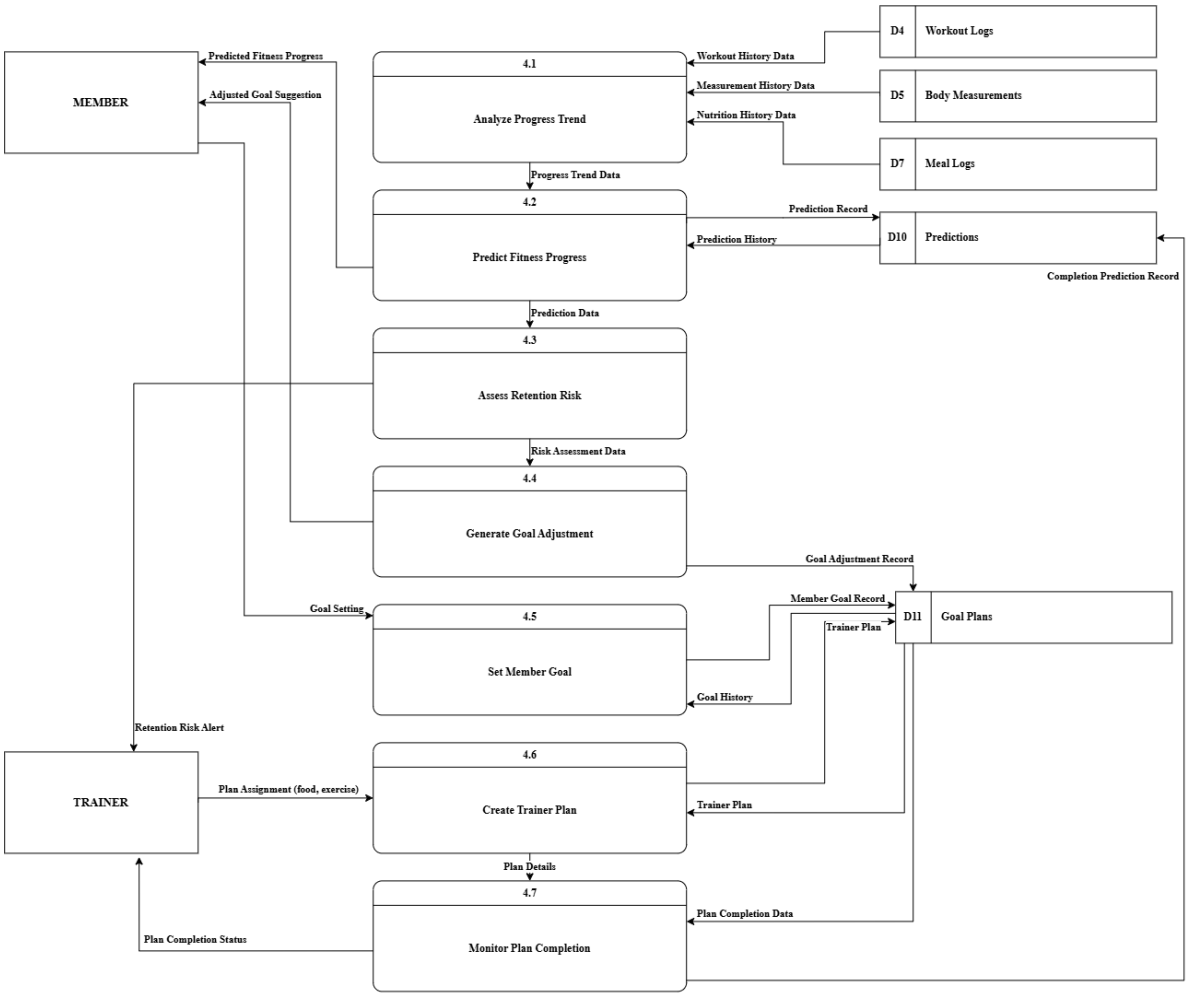


Figure 13: Data Flow Diagram of Process 4 (Proposed)

Figure 13 represents the data flow diagram of process 4 of the proposed system. This diagram decomposes the Generate Predictions and Goal Plans process into seven sub-processes: 4.1 Analyze Progress Trend, 4.2 Predict Fitness Progress, 4.3 Assess Retention Risk, 4.4 Generate Goal Adjustment, 4.5 Set Member Goal, 4.6 Create Trainer Plan, and 4.7 Monitor Plan Completion.

In sub-process 4.1, the workout history data, the measurement history data, and the nutrition history data are retrieved from the Workout Logs, the Body Measurements, and the Meal Logs, and the progress trend data are produced. These data are passed to sub-process4.2,

where the predicted fitness progress is generated and the prediction record is written into the Predictions, while the prediction history is retrieved for comparison.

The prediction data are passed to sub-process 4.3, where the retention risk is assessed and the retention risk alert is returned to the Trainer. The risk assessment data are passed to sub-process 4.4, where the adjusted goal suggestion is generated, returned to the Member, and written into the Goal Plans.

In sub-process 4.5, the Member provides the goal setting and the member goal record is written into the Goal Plans, while the goal history is retrieved. In sub-process 4.6, the Trainer provides the plan assignment containing the food plan and the exercise plan, and the trainer plan record is written into the Goal Plans.

In sub-process 4.7, the plan completion data are retrieved from the Goal Plans and the plan completion status is returned to the Trainer, while the completion prediction record is written into the Predictions.

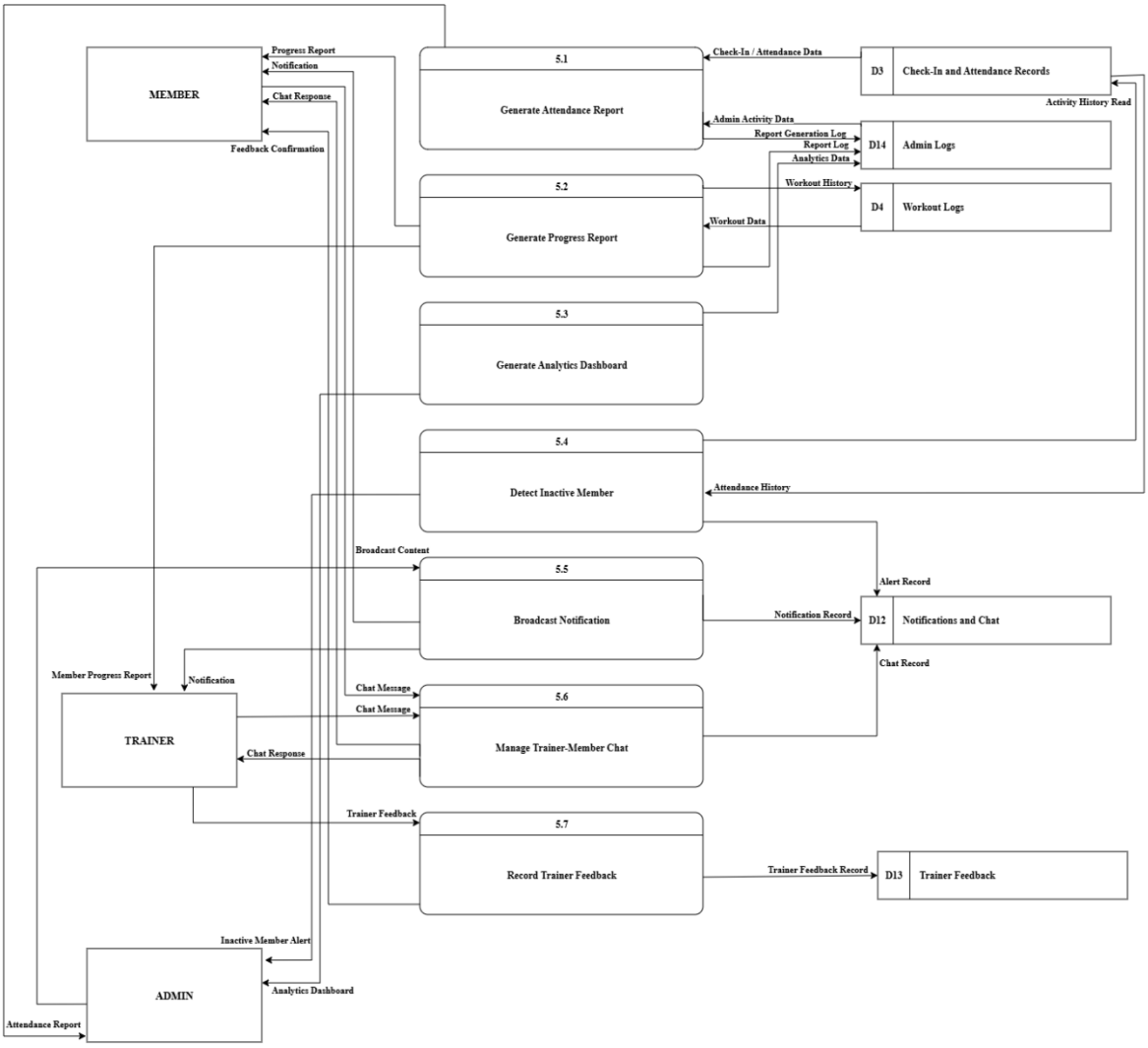


Figure 14: Data Flow Diagram of Process 5 (Proposed)

Figure 14 represents the data flow diagram of process 5 of the proposed system. This diagram decomposes the Generate Reports and Communication process into seven sub-processes: 5.1 Generate Attendance Report, 5.2 Generate Progress Report, 5.3 Generate Analytics Dashboard, 5.4 Detect Inactive Member, 5.5 Broadcast Notification, 5.6 Manage Trainer-Member Chat, and 5.7 Record Trainer Feedback.

In sub-process 5.1, the check-in and attendance data are retrieved from the Check-In and Attendance Records together with the admin activity data from the Admin Logs, and the

attendance report is returned to the Admin while the report generation log is written into the Admin Logs.

In sub-process 5.2, the workout data are retrieved from the Workout Logs to generate the progress report, which is returned to the Member and to the Trainer as the member progress report, while the report log is written into the Admin Logs. In sub-process 5.3, the analytics data log is written into the Admin Logs and the analytics dashboard is returned to the Admin.

In sub-process 5.4, the attendance history is retrieved from the Check-In and Attendance Records to identify members who have stopped attending. The inactive member alert is returned to the Admin and the alert record is written into the Notifications and Chat.

In sub-process 5.5, the Admin provides the broadcast content, which is written into the Notifications and Chat and returned to the Member and the Trainer as a notification. In sub-process 5.6, the chat messages of the Member and the Trainer are exchanged, the chat record is written into the Notifications and Chat, and the chat responses are returned to both parties.

In sub-process 5.7, the Trainer provides the trainer feedback content and the Member provides the trainer rating. The trainer feedback record is written into the Trainer Feedback and the feedback confirmation is returned to the Member.

3.6 ACTIVITY DIAGRAM

An activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. Activity diagrams may be regarded as a form of flowchart.

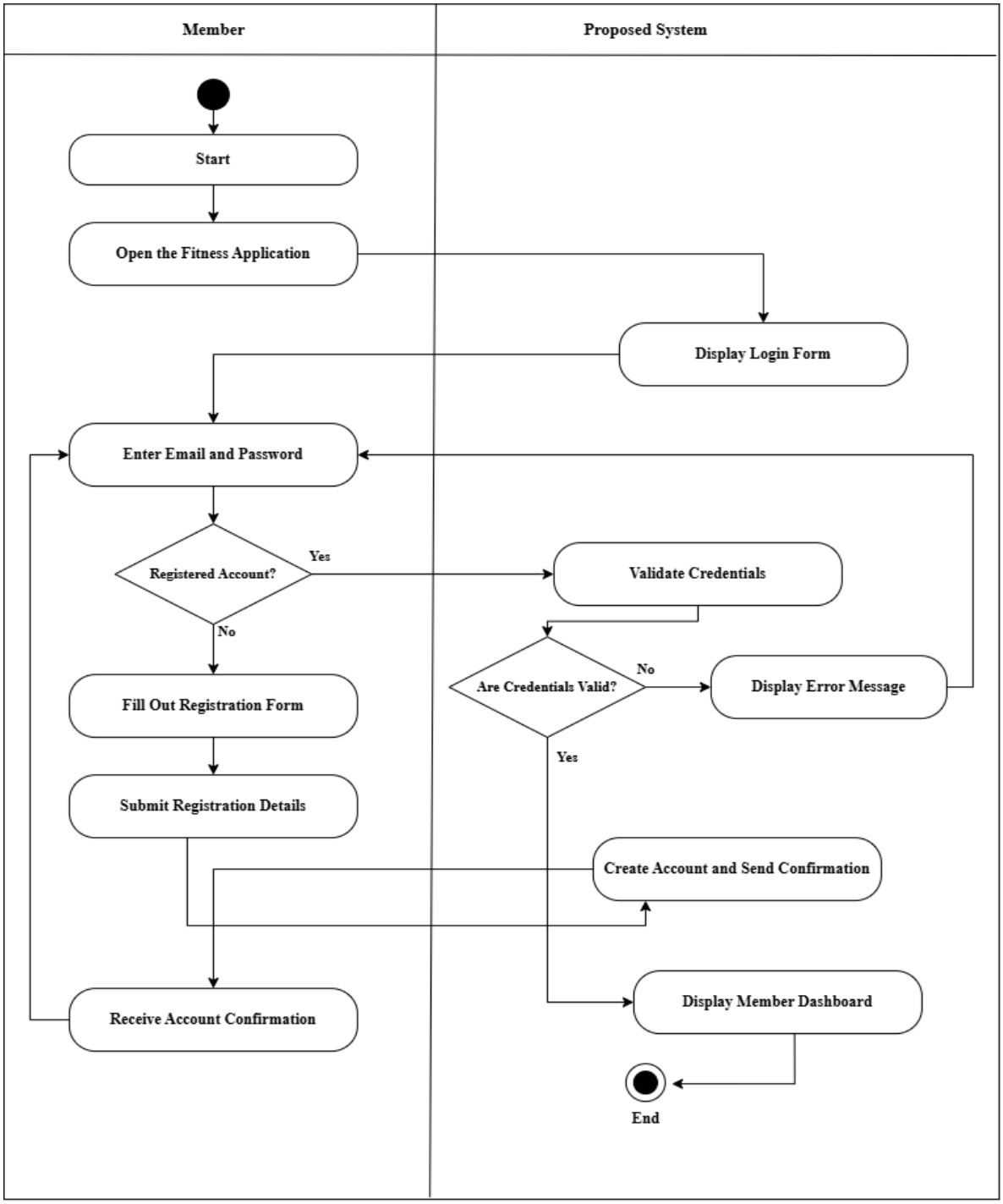


Figure 15: Activity Diagram (Member Login and Registration)

Figure 15 shows the activity diagram of the member login and registration process of the proposed system of Triple J Fitness Center. The process begins when the Member opens the fitness application and enters an email address and a password. The system first determines whether the Member has an existing registered account.

If the Member is not yet registered, the registration form is presented, where the Member fills out the required personal and membership details. The registration details are submitted, the account is created and stored in the User Profiles, and the Member receives an account confirmation before returning to the login step.

If the Member is already registered, the system validates the entered credentials against the stored account records. When the credentials are valid, the system displays the member dashboard and the process ends. When the credentials are invalid, an error message is displayed and the Member is returned to the login step to re-enter the email and password. This automated validation replaces the manual checking of member records in the existing system.

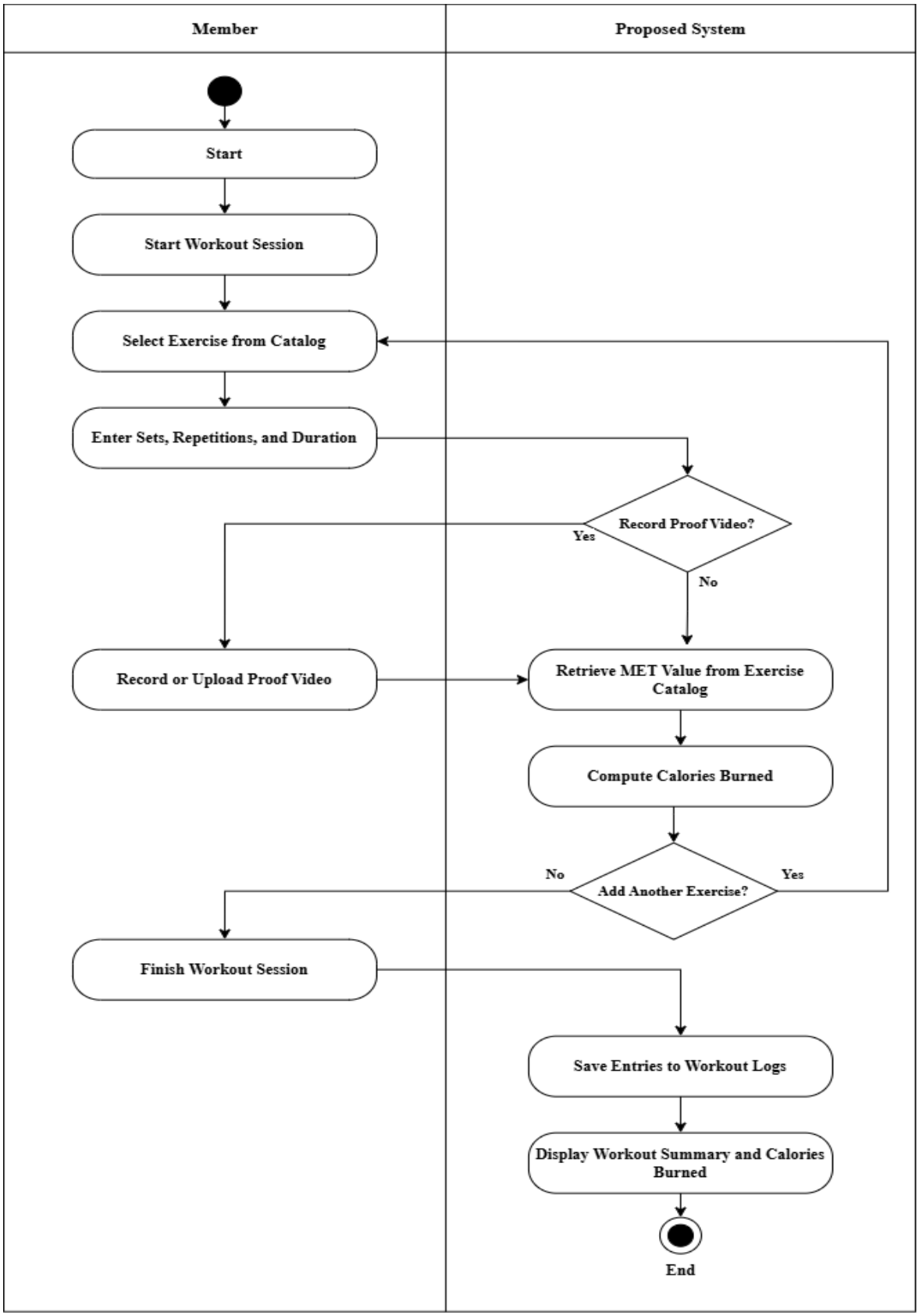


Figure 16: Activity Diagram (Workout Logging with Exercise Tracking)

Figure 16 shows the activity diagram of the workout logging process of the proposed system of Triple J Fitness Center. The process begins when the Member starts a workout session and selects an exercise from the exercise catalog. For each selected exercise, the Member enters the number of sets, the number of repetitions, and the duration of the exercise.

After the exercise details are entered, the system determines whether the Member will record a proof video. If a proof video is recorded, the Member captures or uploads the video before proceeding otherwise, the process continues directly. The system then retrieves the MET value of the selected exercise from the MET Exercise Catalog and computes the calories burned based on the exercise duration and the recorded body weight of the member.

The system then determines whether the Member will add another exercise. If yes, the process returns to the exercise selection step; if no, the Member finishes the session. All recorded entries are saved into the Workout Logs, and the system displays the workout summary together with the total calories burned, which ends the process.

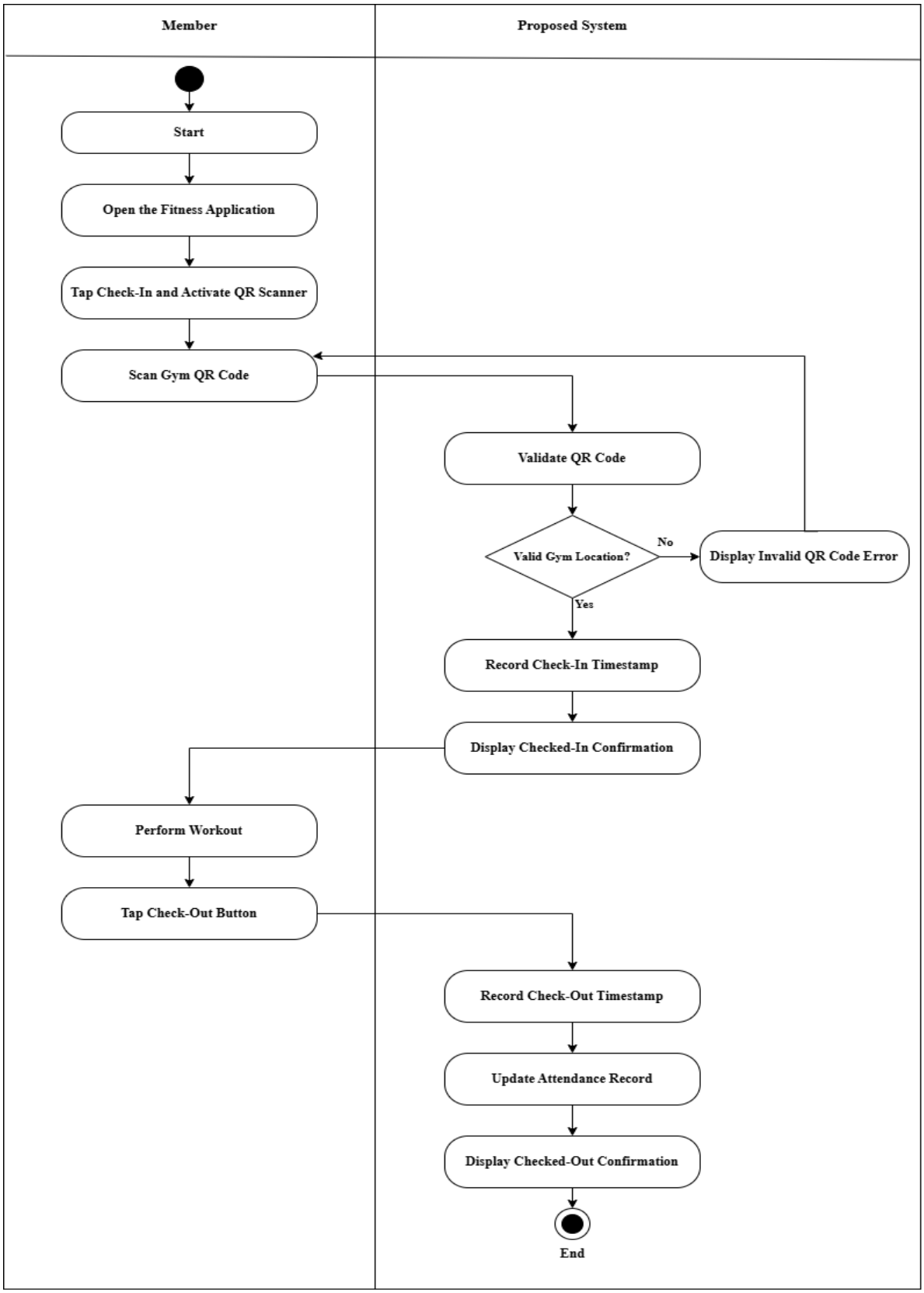


Figure 17: Activity Diagram (QR Code Check-In and Check-Out)

Figure 17 shows the activity diagram of the QR code check-in and check-out process of the proposed system of Triple J Fitness Center. The process begins when the Member opens the application and taps the check-in button, which activates the QR scanner. The Member scans the QR code posted at the gym, and the system validates whether the scanned code corresponds to a recognized gym location.

If the QR code is invalid, an error message is displayed and the Member is returned to the scanning step. If the QR code is valid, the system records the check-in timestamp into the Check-In and Attendance Records, displays a checked-in confirmation, and the Member performs the workout.

After the workout, the Member taps the check-out button. The system records the check-out timestamp, updates the attendance record, and displays a checked-out confirmation, which ends the process. This automated attendance flow replaces the manual log sheets used in the existing system.

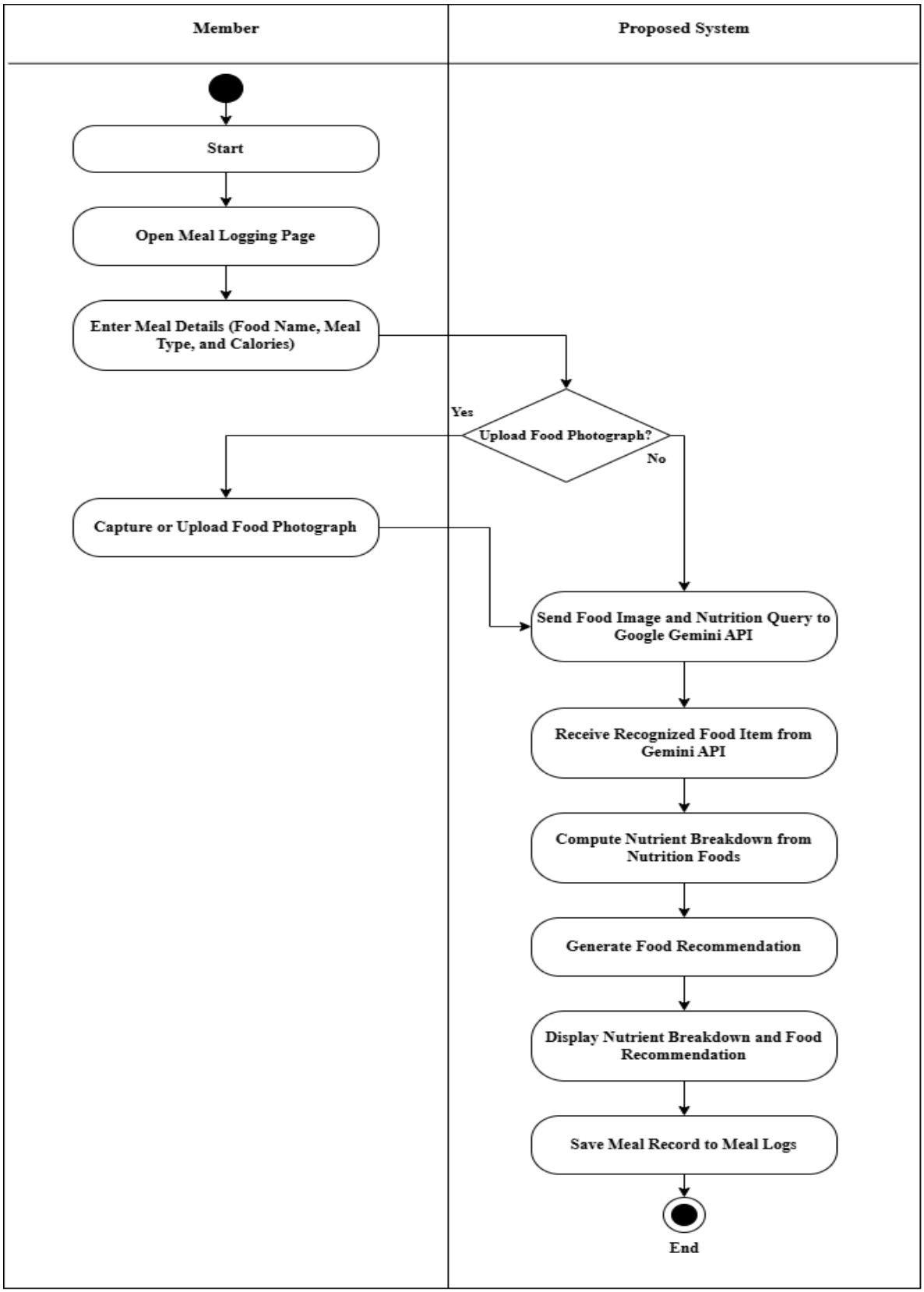


Figure 18: Activity Diagram (Meal Logging and AI Food Analysis)

Figure 18 shows the activity diagram of the meal logging and AI food analysis process of the proposed system of Triple J Fitness Center. The process begins when the Member opens the meal logging page and enters the meal details containing the food name, the meal type, and the estimated calories. The system then determines whether the Member will upload a food photograph.

If a photograph is uploaded, the captured or uploaded image is sent together with a nutrition query to the Google Gemini API, which returns the recognized food item. If no photograph is uploaded, the process proceeds directly using the manually entered meal details.

The system computes the nutrient breakdown by retrieving the food composition data from the Nutrition Foods, generates a food recommendation, and displays the nutrient breakdown together with the recommendation to the Member. Finally, the meal record is saved into the Meal Logs, which ends the process.

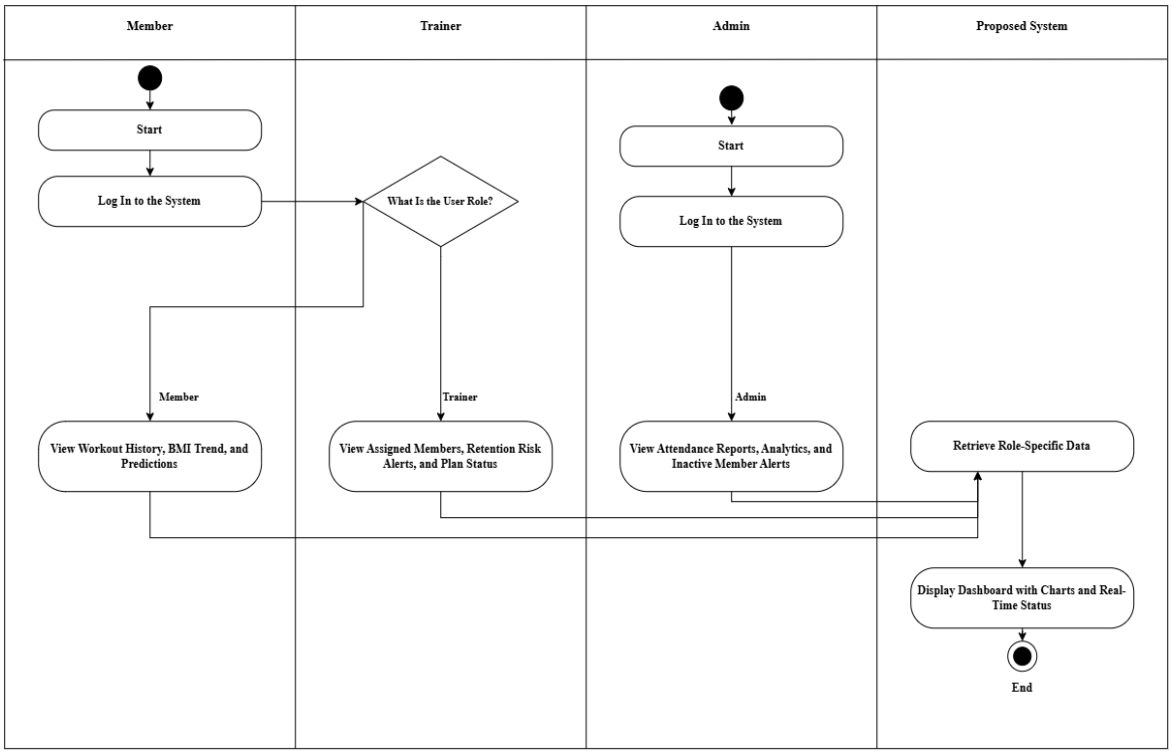


Figure 19: Activity Diagram (Dashboard View)

Figure 19 shows the activity diagram of the dashboard view process of the proposed system of Triple J Fitness Center. The process begins when the user logs in to the system, and the system determines the role of the logged-in user.

When the user is a Member, the dashboard presents the workout history, the BMI trend, and the predicted fitness progress. When the user is a Trainer, the dashboard presents the assigned members, the retention risk alerts, and the plan completion status. When the user is an Admin, the dashboard presents the attendance reports, the analytics summary, and the inactive member alerts.

After the role-specific selections are made, the system retrieves the corresponding data from the Workout Logs, the Body Measurements, the Predictions, the Goal Plans, the Check-In and Attendance Records, and the Admin Logs, and displays the dashboard containing the charts and the real-time status, which ends the process.

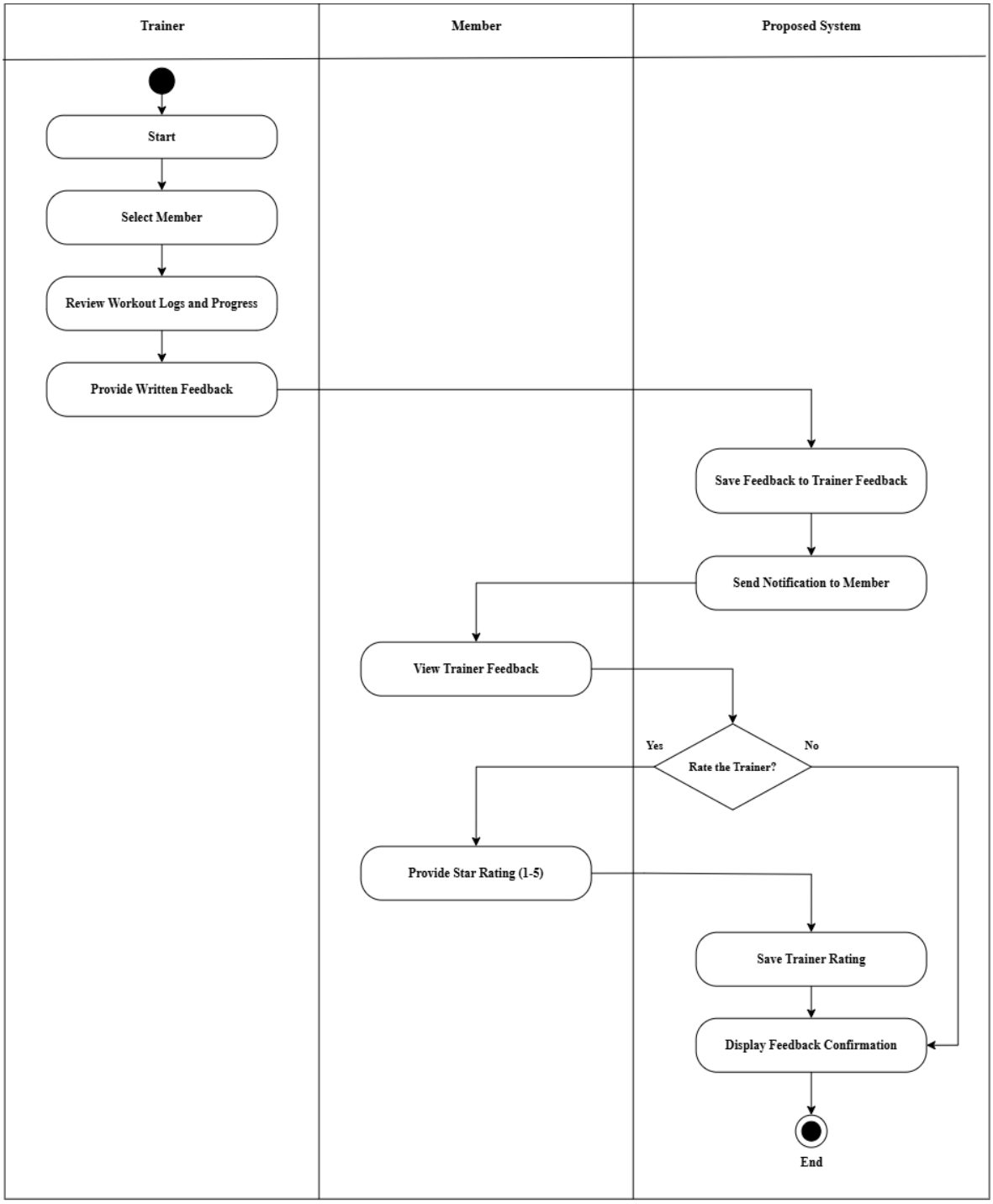


Figure 20: Activity Diagram (Trainer Feedback and Member Rating)

Figure 20 shows the activity diagram of the trainer feedback and member rating process of the proposed system of Triple J Fitness Center. The process begins when the Trainer selects a member and reviews the workout logs and progress of that member. The Trainer provides written feedback, which the system saves into the Trainer Feedback, and the system sends a notification to the Member.

The Member views the feedback and chooses whether to rate the trainer. If the Member provides a star rating from one to five, the system saves the rating; otherwise the process proceeds directly. The system then displays the feedback confirmation to the Member, which ends the process. This closed-loop feedback mechanism replaces the informal verbal advice of the existing system with recorded and rated feedback.

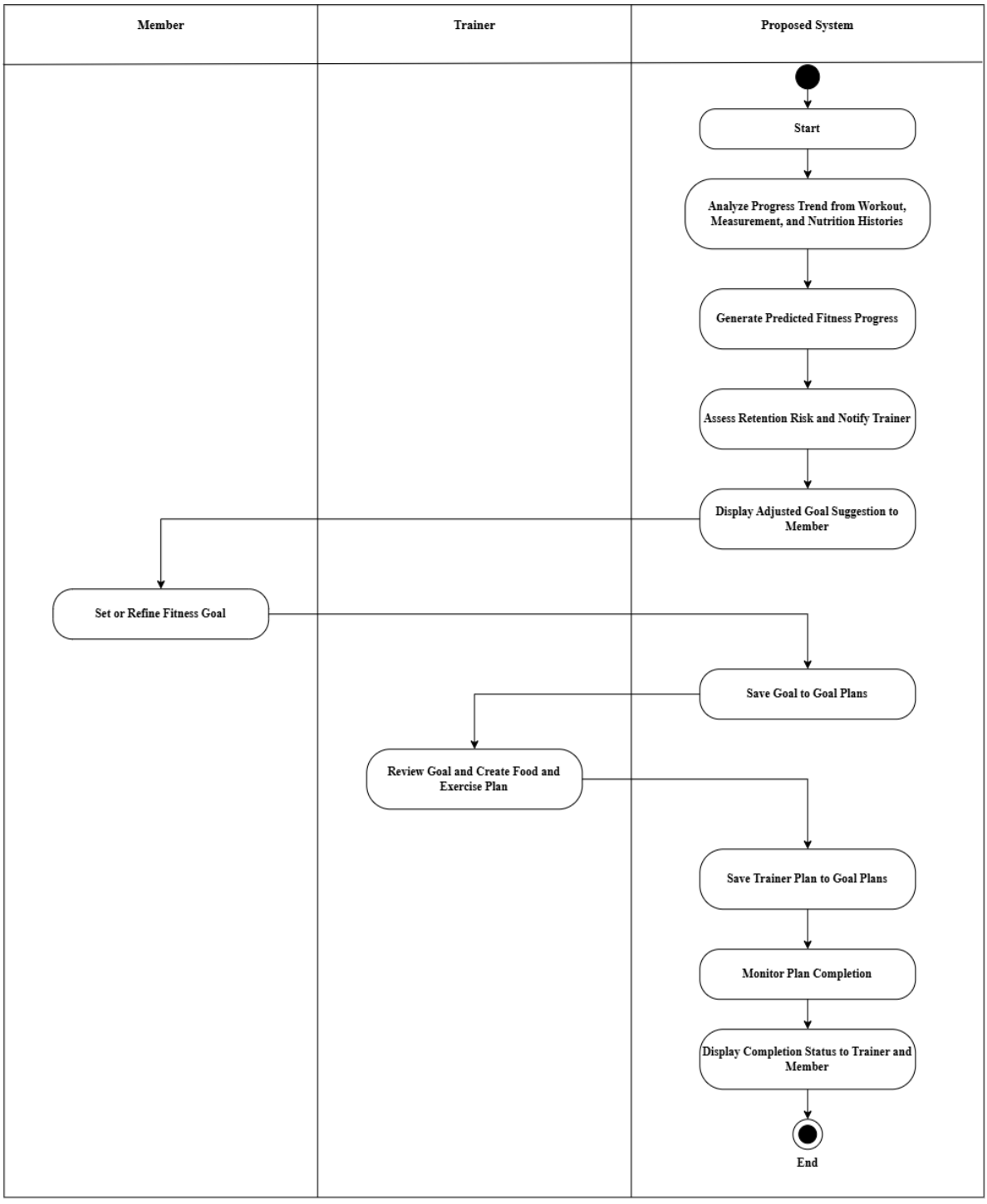


Figure 21: Activity Diagram (Goal Setting and Plan Creation)

Figure 21 shows the activity diagram of the goal setting and plan creation process of the proposed system of Triple J Fitness Center. The process begins when the system analyzes the progress trend of the member using the workout history, the measurement history, and the

nutrition history retrieved from the Workout Logs, the Body Measurements, and the Meal Logs. Using the analyzed trend, the system generates the predicted fitness progress, assesses the retention risk, and notifies the Trainer of the risk result. The adjusted goal suggestion is displayed to the Member, who sets or refines the fitness goal, and the goal is saved into the Goal Plans.

The Trainer reviews the member goal and creates the plan consisting of the food plan and the exercise plan, which is saved into the Goal Plans. The system continuously monitors the plan completion and displays the completion status to the Trainer and the Member, which ends the process.

3.7 ENTITY RELATIONSHIP DIAGRAM

An entity–relationship model (ER model) is a data model for describing the data or information aspects of a business domain or its process requirements, in an abstract way that lends itself to ultimately being implemented in a database such as a relational database. The main components of ER models are entities and the relationships that can exist among them, and databases.

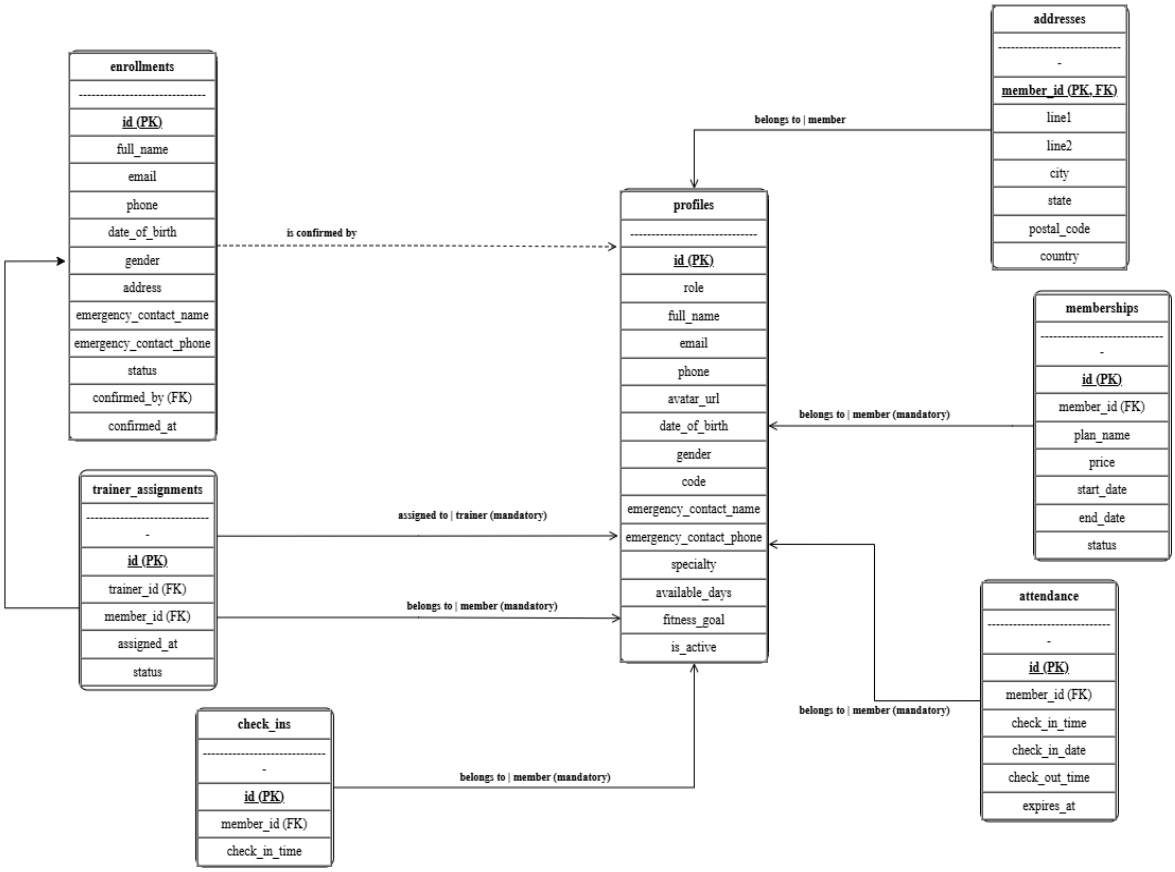


Figure 22: Entity Relationship Diagram (Accounts, Membership and Attendance)

Figure 22 presents the accounts, membership and attendance part of the entity relationship diagram of the proposed system. The profiles entity is the central entity because it stores the account records of the Admin, the Trainer, and the Member. The enrollments entity stores the pending registration details and refers to profiles through the confirmed by foreign key, which is optional because an enrollment isconfirmed only after the Admin approves it. The memberships entity stores the plan name, the price, and the validity period of each membership. The trainer assignments entity links each member to an assigned trainer through both the trainer

and the member foreign keys, while the attendance and check in entities store the QR code check-in and check-out records, and the addresses entity stores the home address of each member.

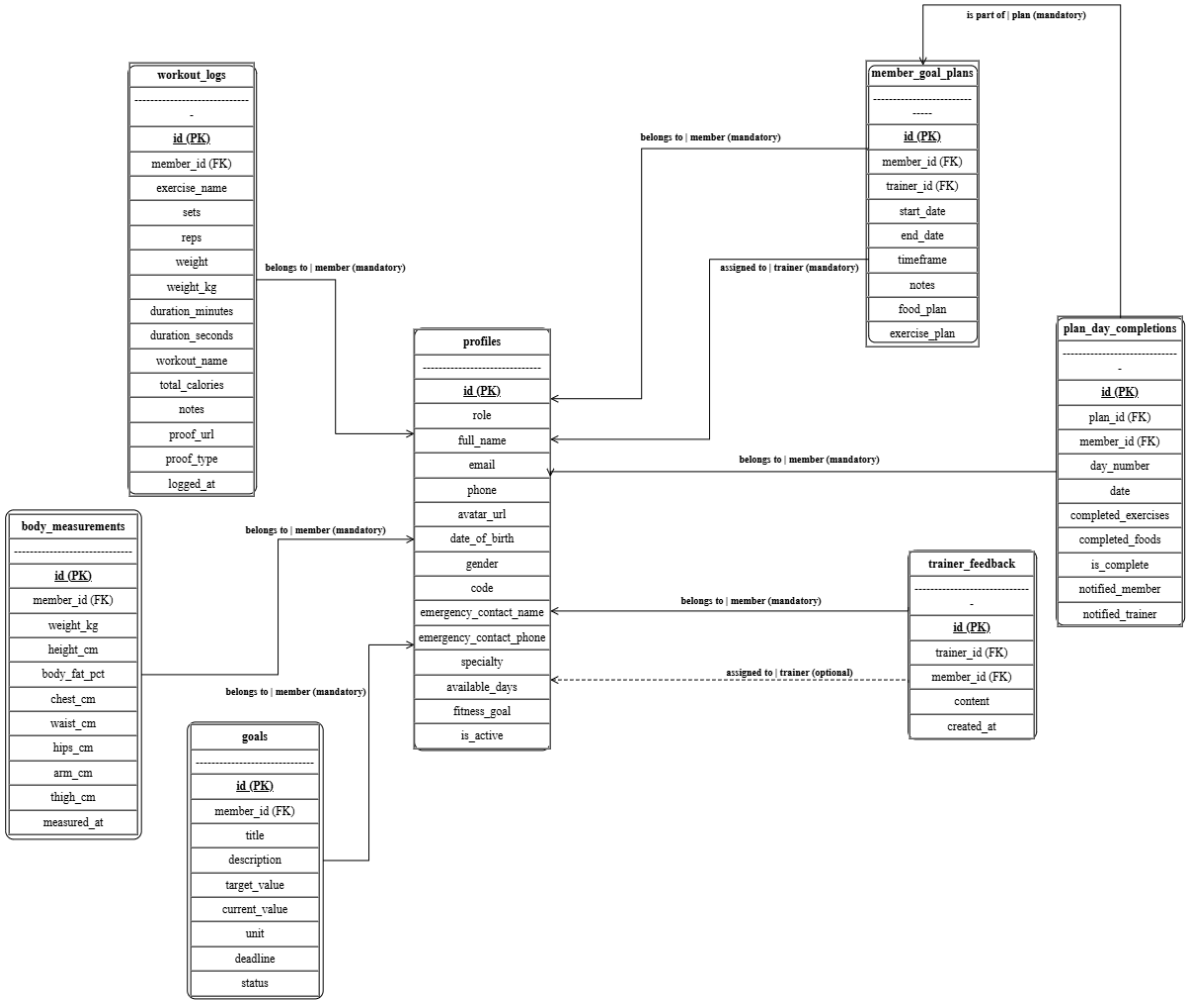


Figure 23: Entity Relationship Diagram (Fitness Tracking, Goals and Trainer Feedback)

Figure 23 presents the fitness tracking, goals, and trainer feedback part of the entity relationship diagram. The workout_logs entity stores each exercise entry together with the sets, the repetitions, the weight, the duration, the proof video, and the computed calories; the body_measurements entity stores the weight, the height, and the body circumference readings; and the goals entity stores the fitness goals set by the member. The member_goal_plans entity stores the seven-day food plan and exercise plan assigned by the trainer, and the

plan_day_completions entity records the per-day completion state of each plan, referring both to member_goal_plans through the plan_id foreign key and to profiles through the member_id foreign key. The trainer_feedback entity stores the written feedback of the trainer for each member; its trainer_id foreign key is optional because feedback is written only by trainer accounts.

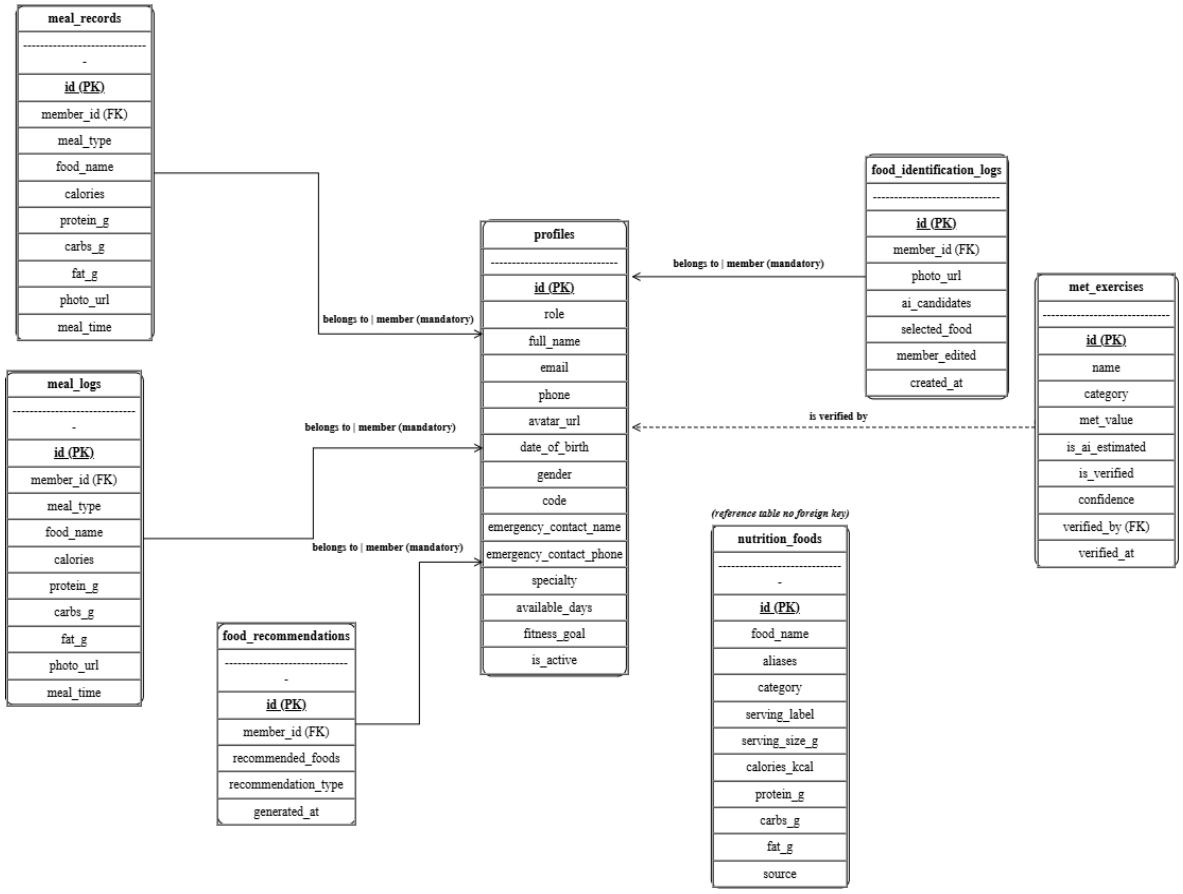


Figure 24: Entity Relationship Diagram (Nutrition, Meal Logging and Reference Data)

Figure 24 presents the nutrition, meal logging, and reference data part of the entity relationship diagram. The meal_records and meal_logs entities store the logged meals and their nutrient values; the food_recommendations entity stores the generated food suggestions; and the food_identification_logs entity stores the AI photo recognition candidates and the member's selected correction. The met_exercises reference entity stores the exercise catalog together with the MET value of each exercise and the optional verifier through the verified_by foreign key. The nutrition_foods reference entity stores the verified Filipino dish composition data taken

from the DOST-FNRI reference; it has no foreign key to profiles and is therefore shown as a standalone table in the figure.

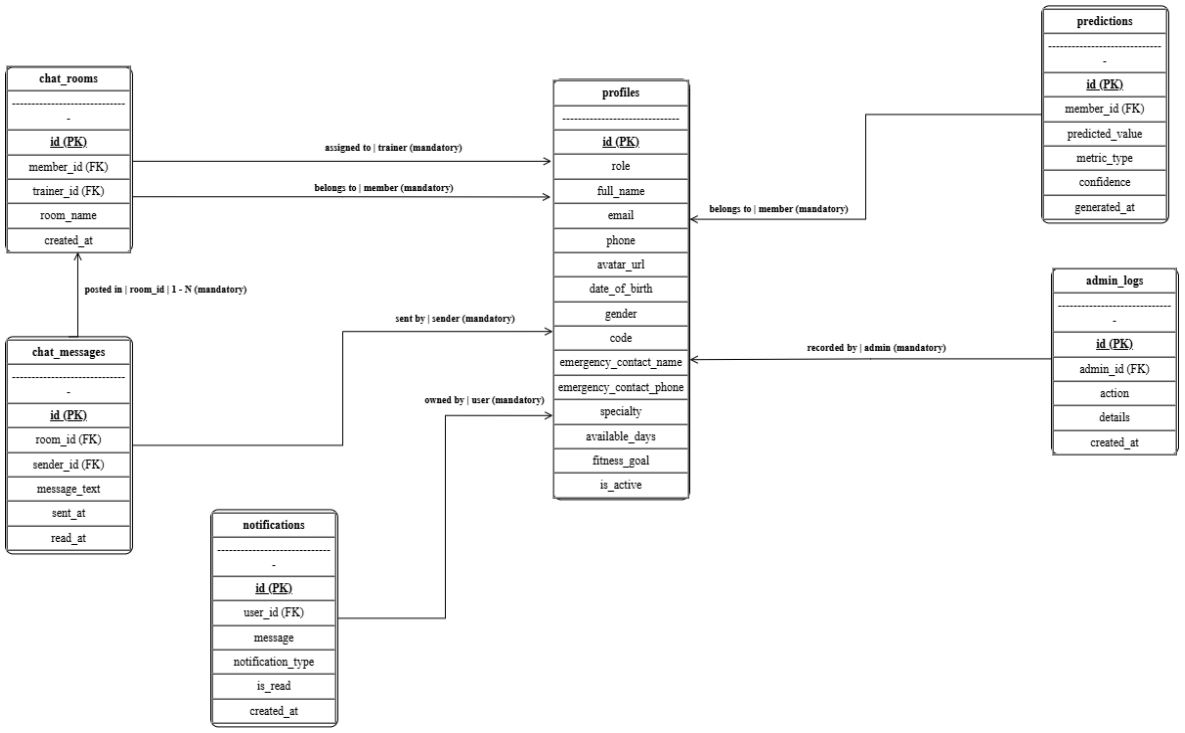


Figure 25: Entity Relationship Diagram (Communication, Notifications and Administration)

Figure 25 presents the communication, notifications, and administration part of the entity relationship diagram. The chat_rooms entity stores each trainer-member conversation room through the member_id and trainer_id foreign keys, while chat_messages stores the messages of each room through room_id and sender_id. The notifications entity stores system alerts, the predictions entity stores generated fitness progress predictions, and the admin_logs entity stores administrator activity through admin_id.

Across Figures 22 to 25, the profiles entity is repeated as the central entity because it stores the account records of the Admin, Trainer, and Member. The three optional relationships are enrollments to profiles through confirmed_by, trainer_feedback to profiles through trainer_id, and met_exercises to profiles through verified_by, while the other relationships are mandatory one-to-many relationships shown with crow's foot notation.

3.8 USE CASE DIAGRAM

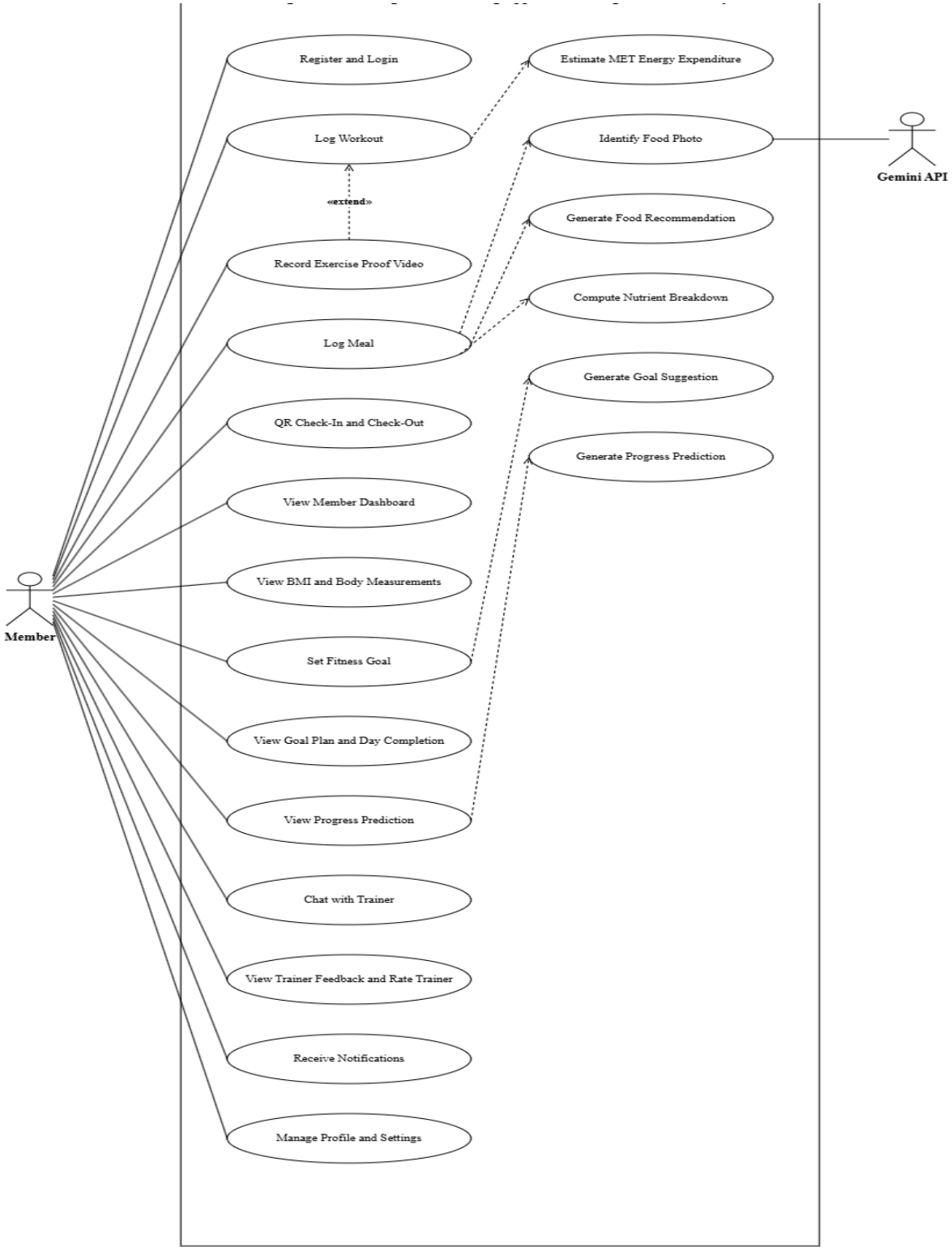


Figure 22: Entity Relationship Diagram (Accounts, Membership and Attendance)

Figure 22 presents the accounts, membership and attendance part of the entity relationship diagram of the proposed system. The profiles entity is the central entity because it stores the account records of the Admin, the Trainer, and the Member. The enrollments entity stores the pending registration details and refers to profiles through the confirmed_by foreign key, which is optional because an enrollment is confirmed only after the Admin approves it. The memberships entity stores the plan name, the price, and the validity period of eachmembership. The trainer_assignments entity links each member to an assigned trainer through both the trainer_id and the member_id foreignkeys, while the attendance and check_ins entities store the QR code check-in and check-out records, and the addresses entity stores thehome address of each member.



Figure 27: Use Case of The Proposed System

Figure 27 Use Case of the Proposed System represents the access rights of the **Trainer**. The Trainer can access different functions of the system, including logging in, viewing the trainer dashboard with retention risk, viewing assigned members and member logs, recording body measurements, creating a 7-day food and exercise plan, monitoring plan completion, giving feedback, chatting with members, receiving notifications, and managing their profile. The Trainer can also assess retention risk, generate goal suggestions, and search the MET exercise catalog to support the fitness plans of assigned members.

The Trainer can use the Assess Retention Risk function to view and evaluate the possible retention risk of members based on the information available in the system. The Trainer can also generate goal suggestions and search the MET Exercise Catalog when creating personalized food and exercise plans for members. These functions help the Trainer monitor member progress and provide appropriate fitness guidance based on the member's information and goals.

The Trainer can also monitor the completion of the assigned 7-day food and exercise plan and provide feedback to members. Through the chat function, the Trainer can communicate directly with members, while notifications allow the Trainer to receive important system updates. The Trainer can also manage their profile and access the functions provided according to their role in the system.

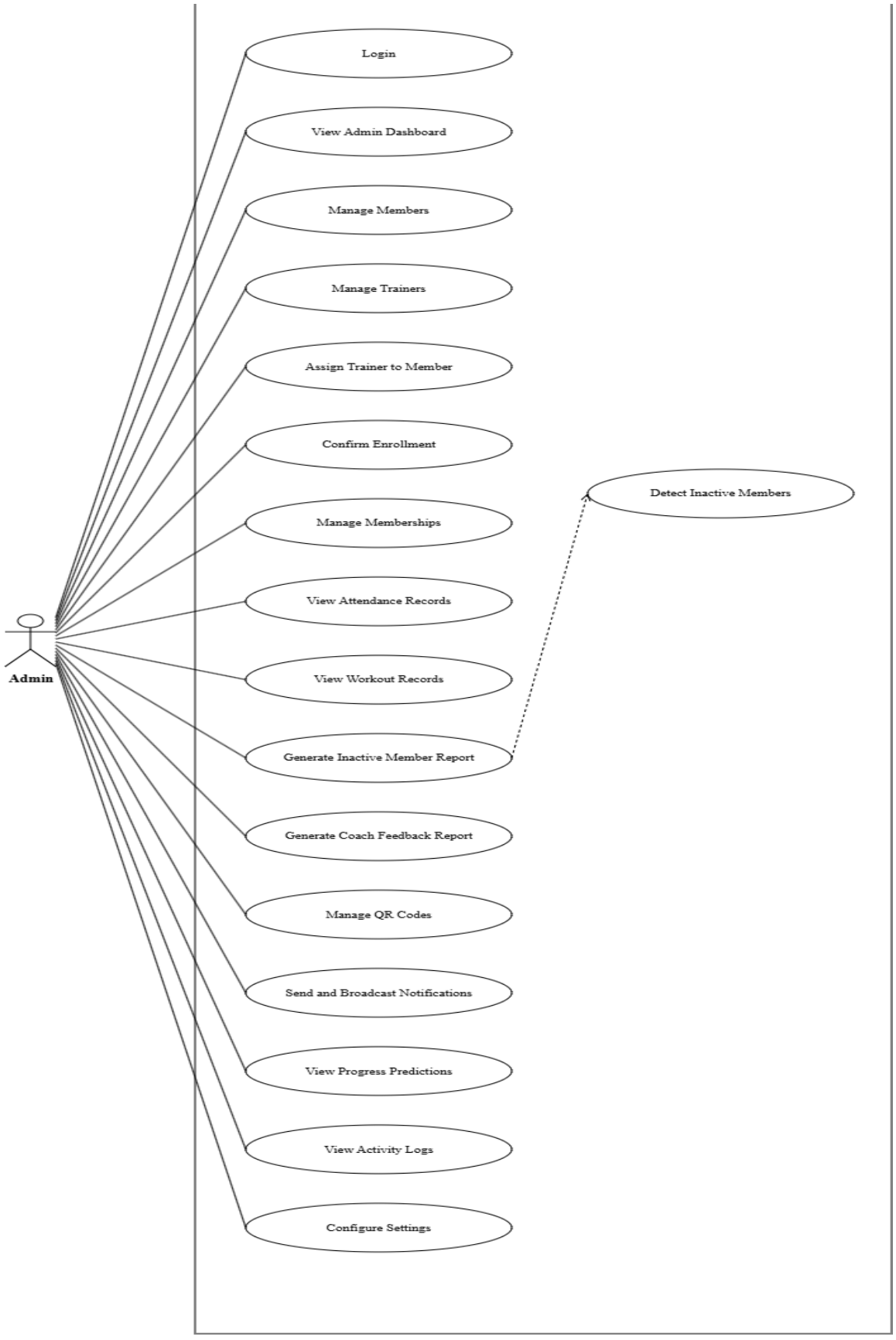

Login
View Admin Dashboard
Manage Members
Manage Trainers
Assign Trainer to Member
Confirm Enrollment
Manage Memberships
View Attendance Records
View Workout Records
Generate Inactive Member Report
Generate Coach Feedback Report
Manage QR Codes
Send and Broadcast Notifications
View Progress Predictions
View Activity Logs
Configure Settings
Detect Inactive Members
Admin

Figure 25: Entity Relationship Diagram (Communication, Notifications and Administration)

Figure 25 presents the communication, notifications, and administration part of the entity relationship diagram. The chat_rooms entity stores each trainer-member conversation room through both the member_id and the trainer_id foreign keys, and the chat_messages entity stores the messages of each room, referring to chat_rooms through the room_id foreign key and to profiles through the sender_id foreign key. The notifications entity stores the system alerts sent to each account, the predictions entity stores the generated fitness progress predictions, and the admin_logs entity stores the activity log of the Admin through the admin_id foreign key.

Across Figures 22 to 25 the profiles entity is repeated as the central entity because it stores the account records of the Admin, the Trainer, and the Member, and almost every other entity refers to it through a foreign key. The three optional relationships in the diagram are enrollments to profiles through confirmed_by, trainer_feedback to profiles through trainer_id, and met_exercises to profiles through verified_by; all other relationships are mandatory one-to-many relationships shown with crow's foot notation.

3.9 DATA DICTIONARY

These tables present the tables used in the system and defines what data should be stored in each table.

Table 3: Enrollments

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Enrollments table.	uuid	--	PK
full_name	Full name of the enrolling prospect or member.	text	--	
email	Email address used for communication and account identification.	text	--	
phone	Phone number for SMS or voice contact.	text	--	
date_of_birth	Date of birth used for age-appropriate program recommendations.	date	--	
gender	Text indicating the enrollee's gender.	text	--	
address	Text field capturing the enrollee's home address.	text	--	
emergency_contact_name	Name of the designated emergency contact person.	text	--	
emergency_contact_phone	Phone number of the designated emergency contact.	text	--	

emergency_contact_relation	Text describing the relationship to the emergency contact.	text	--	
status	Text indicating the enrollment's current processing status.	text	--	
confirmed_at	Timestamp when the enrollment was confirmed.	time stamp	--	
confirmed_by	Foreign key referencing the profile of the user who confirmed the enrollment in the Enrollments table.	uuid	--	FK
created_at	Timestamp recording when the record was created in the Enrollments table.	time stamp	--	
state_updated_at	Timestamp with time zone stored in the Enrollments table.	time stamp	--	

The Enrollments table captures prospect and member registration details, including contact information, demographics, and membership confirmation tracking.

Table 4: Profiles

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Profiles table.	uuid	--	PK
role	Enumeration value (admin, trainer, or member) defining the profile's access role.	user_role	--	
full_name	Text field stored in the Profiles table.	text	--	
email	Text field stored in the Profiles table.	text	--	
phone	Text field stored in the Profiles table.	text	--	
avatar_url	URL pointing to the profile's avatar image.	text	--	
date_of_birth	Date value stored in the Profiles table.	date	--	
gender	Text field stored in the Profiles table.	text	--	
created_at	Timestamp recording when the record was created in the Profiles table.	time stamp	--	
updated_at	Timestamp recording the last update to the record in the Profiles table.	time stamp	--	

fitness_goal	Text describing the member's overall fitness goal or focus area.	text	--	
code	Text identifier or referral code associated with the profile.	text	--	
emergency_contact_name	Text field stored in the Profiles table.	text	--	
emergency_contact_phone	Text field stored in the Profiles table.	text	--	
specialty	Text describing the trainer's area of specialization.	text	--	
available_days	Text listing the days the trainer is available for sessions.	text	--	
is_active	Boolean flag indicating whether the profile is currently active.	boolean	--	

The Profiles table stores authenticated user accounts for members, trainers, and administrators, recording role assignment, profile attributes, and account status.

Table 5: Memberships

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Memberships table.	uuid	--	PK
member_id	Foreign key referencing the member's profile.	uuid	--	FK
plan_name	Text name of the membership plan purchased.	text	--	
price	Decimal price of the membership plan.	decimal	10,2	
start_date	Date the membership period began.	date	--	
end_date	Date the membership period ends.	date	--	
status	Enumeration value (active, expired, or cancelled) indicating membership state.	membership_status	--	
created_at	Timestamp recording when the record was created in the Memberships table.	time stamp	--	
updated_at	Timestamp recording the last update to the record in the Memberships table.	time stamp	--	

The Memberships table tracks each member's active or historical membership plans, including pricing, billing period, and current membership status.

Table 6: Trainer Assignments

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Trainer Assignments table.	uuid	--	PK
trainer_id	Foreign key referencing the assigned trainer's profile.	uuid	--	FK
member_id	Foreign key referencing the assigned member's profile.	uuid	--	FK
assigned_at	Timestamp when the trainer-member assignment was created.	time stamp	--	
status	Enumeration value (active or ended) indicating assignment state.	assignment_status	--	

The Trainer Assignments table records the association between a trainer and a member, including the assignment date and current assignment status.

Table 7: Attendance

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Attendance table.	uuid	--	PK
member_id	Foreign key referencing the member's profile in the Attendance table.	uuid	--	FK
check_in_time	Timestamp when the member checked in for the session.	time stamp	--	
check_in_date	Date of the attended session.	date	--	
check_out_time	Timestamp when the member checked out of the session.	time stamp	--	
expires_at	Timestamp after which the attendance record is considered stale.	time stamp	--	

The Attendance table logs each member's gym visit sessions by recording check-in and check-out times, the visit date, and session expiry.

Table 8: Check-Ins

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Check-Ins table.	uuid	--	PK
member_id	Foreign key referencing the member's profile in the Check-Ins table.	uuid	--	FK
check_in_time	Timestamp recording when the member entered the gym.	time stamp	--	

The Check-Ins table records the timestamp of each member's gym entry event for quick presence tracking.

Table 9: Addresses

Field Name	Description	Field Type	Field Size	Type of Key
member_id	Primary key and foreign key referencing the member's profile; each member has one address row.	uuid	--	PK
line1	First line of the street address.	text	--	
line2	Optional second line of the street address.	text	--	
city	City name of the address.	text	--	
state	State or province of the address.	text	--	
postal_code	Postal or ZIP code of the address.	text	--	
country	Country of the address.	text	--	

created_at	Timestamp recording when the record was created in the Addresses table.	time stamp	--	
updated_at	Timestamp recording the last update to the record in the Addresses table.	times stamp	--	

The Addresses table stores the physical mailing address associated with each member, keyed directly to the member's profile.

Table 10: Workout Logs

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Workout Logs table.	uuid	--	PK
member_id	Foreign key referencing the member who performed the exercise.	uuid	--	FK
exercise_name	Text name of the exercise performed.	text	--	
sets	Integer value stored in the Workout Logs table.	int	--	
reps	Integer count of repetitions performed.	int	--	
weight	Decimal weight value in the legacy field (superseded by weight_kg).	decimal	10,2	
duration_minutes	Decimal duration in minutes in the legacy field (superseded by duration_seconds).	int	--	
notes	Text notes about the exercise set.	text	--	
logged_at	Timestamp when the workout log entry was recorded.	time stamp	--	

proof_url	Text field stored in the Workout Logs table.	text	--	
proof_type	Text field stored in the Workout Logs table.	text	--	
weight_kg	Decimal weight in kilograms used for the set.	decimal	10,2	
duration_seconds	Decimal duration in seconds for the exercise set.	int	--	
workout_name	Text field stored in the Workout Logs table.	text	--	
total_calories	Integer value stored in the Workout Logs table.	integer	--	

The Workout Logs table records individual exercise sessions performed by a member, capturing exercise type, sets, repetitions, and load in both metric and legacy fields.

Table 11: Goals

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Goals table.	uuid	--	PK
member_id	Foreign key referencing the member who owns the goal.	uuid	--	FK
title	Short text title of the fitness goal.	text	--	
description	Text describing the goal in detail.	text	--	

target_value	Decimal target value the member aims to reach.	decimal	10,2	
current_value	Decimal current progress value toward the target.	decimal	10,2	
unit	Text unit of measure for the target and current values.	text	--	
deadline	Date by which the goal is targeted to be achieved.	date	--	
status	Text indicating the goal's current state (for example, active or achieved).	text	--	
created_at	Timestamp recording when the record was created in the Goals table.	time stamp	--	
updated_at	Timestamp recording the last update to the record in the Goals table.	time stamp	--	

The Goals table defines a member's fitness objectives with a target value, current progress, unit of measure, deadline, and status.

Table 12: Member Goal Plans

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Member Goal Plans table.	uuid	--	PK
member_id	Foreign key referencing the member assigned to the plan.	uuid	--	FK
trainer_id	Foreign key referencing the trainer who created the plan.	uuid	--	FK
start_date	Date the goal plan period begins.	date	--	
end_date	Date the goal plan period ends.	date	--	
timeframe	Text describing the plan's timeframe category.	text	--	
notes	Text notes from the trainer about the plan.	text	--	
food_plan	JSON object containing the prescribed food plan.	jsonb	--	
exercise_plan	JSON object containing the prescribed exercise plan.	jsonb	--	
created_at	Timestamp recording when the record was created in the Member Goal Plans table.	time stamp	--	
updated_at	Timestamp recording the last update to the record in the Member Goal Plans table.	time stamp	--	

The Member Goal Plans table links a member and trainer to a structured plan window, storing timeframe, notes, and JSON-formatted food and exercise plans.

Table 13: Plan Day Completions

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Plan Day Completions table.	uuid	--	PK
plan_id	Foreign key referencing the member goal plan being completed.	uuid	--	FK
member_id	Foreign key referencing the member completing the day.	uuid	--	FK
day_number	Integer day number within the goal plan.	int	--	
date	Date the completion entry corresponds to.	date	--	
completed_exercises	JSON object listing the exercises completed that day.	json	--	
completed_foods	JSON object listing the foods completed that day.	json	--	
is_complete	Boolean indicating whether the day's plan was fully completed.	boolean	--	
notified_member	Boolean indicating whether the member was notified.	boolean	--	
notified_trainer	Boolean indicating whether the trainer was notified.	boolean	--	

created_at	Timestamp recording when the record was created in the Plan Day Completions table.	time stamp	--	
updated_at	Timestamp recording the last update to the record in the Plan Day Completions table.	time stamp	--	

The Plan Day Completions table records a member's daily progress against a goal plan, including completed exercises, completed foods, and notification state.

Table 14: Trainer Feedback

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Trainer Feedback table.	uuid	--	PK
trainer_id	Foreign key referencing the trainer who authored the feedback.	uuid	--	FK
member_id	Foreign key referencing the member receiving the feedback.	uuid	--	FK
content	Text body of the trainer's feedback to the member.	text	--	
created_at	Timestamp recording when the record was created in the Trainer Feedback table.	time stamp	--	

The Trainer Feedback table stores written feedback entries submitted by a trainer to a member.

Table 15: Meal Records

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Meal Records table.	uuid	--	PK
member_id	Foreign key referencing the member who logged the meal.	uuid	--	FK
meal_type	Text categorizing the meal (for example, breakfast or lunch).	text	--	
food_items	Text listing the food items in the meal.	text	--	
calories	Integer total calorie count for the meal.	int	--	
protein_g	Decimal protein amount in grams.	decimal	6,2	
carbs_g	Decimal carbohydrate amount in grams.	decimal	6,2	
fat_g	Decimal fat amount in grams.	decimal	6,2	
recorded_at	Timestamp when the meal record was logged.	time stamp	--	

The Meal Records table logs a member's self-reported meals with aggregated macronutrient totals and calorie count.

Table 16: Meal Logs

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Meal Logs table.	uuid	--	PK
member_id	Foreign key referencing the member who logged the meal.	uuid	--	FK
meal_type	Text categorizing the meal (for example, breakfast or lunch).	text	--	
food_name	Text name of the recognized food item.	text	--	
calories	Integer calorie count for the food item.	int	--	
protein_g	Decimal protein amount in grams.	decimal	6,2	
carbs_g	Decimal carbohydrate amount in grams.	decimal	6,2	
fat_g	Decimal fat amount in grams.	decimal	6,2	
photo_url	URL of the photo used to recognize the meal.	text	--	
meal_time	Timestamp when the meal was consumed.	time stamp	--	

The Meal Logs table logs an individual recognized meal entry with food name, macronutrients, optional photo, and meal timestamp.

Table 17: Food Recommendations

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Food Recommendations table.	uuid	--	PK
member_id	Foreign key referencing the member receiving the recommendation.	uuid	--	FK
meal_type	Text categorizing the recommended meal type.	text	--	
food_name	Text name of the recommended food.	text	--	
portion_size	Text describing the recommended portion size.	text	--	
reason	Text explaining why the food was recommended.	text	--	
created_at	Timestamp recording when the record was created in the Food Recommendations table.	time stamp	--	

The Food Recommendations table stores AI-generated food suggestions delivered to a member, including meal type, portion, and reasoning.

Table 18: Food Identification Logs

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Food Identification Logs table.	uuid	--	PK
member_id	Foreign key referencing the member who submitted the photo.	uuid	--	FK
photo_url	URL of the uploaded food photo.	text	--	
ai_candidates	JSON array of AI-generated food candidate objects.	json	--	
selected_food	Text name of the food ultimately selected.	text	--	
member_edited	Boolean indicating whether the member edited the AI suggestion.	boolean	--	
created_at	Timestamp recording when the record was created in the Food Identification Logs table.	time stamp	--	

The Food Identification Logs table records each AI food-recognition attempt from a member's photo, including candidate list, selected food, and whether the member edited the result.

Table 19: Nutrition Foods

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Nutrition Foods table.	uuid	--	PK
food_name	Text name of the food in the reference catalog.	text	--	
aliases	Array of text synonyms or alternate names for the food.	text	--	
category	Text category classifying the food (for example, protein or grain).	text	--	
serving_label	Text label describing the standard serving (for example, 1 cup).	text	--	
serving_size_g	Numeric serving size in grams.	numeric	--	
calories_kcal	Numeric calorie count per serving in kilocalories.	numeric	--	
protein_g	Numeric protein amount per serving in grams.	numeric	--	
carbs_g	Numeric carbohydrate amount per serving in grams.	numeric	--	
fat_g	Numeric fat amount per serving in grams.	numeric	--	
source	Text citing the data source for the nutritional values.	text	--	
created_at	Timestamp recording when the record was created in the Nutrition Foods table.	time stamp	--	

The Nutrition Foods table is a reference catalog of foods with serving size, caloric and macronutrient values, category, aliases, and sourcing metadata.

Table 20: MET Exercises

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the MET Exercises table.	uuid	--	PK
name	Text name of the exercise in the reference catalog.	text	--	
category	Text category classifying the exercise.	text	--	
met_value	Numeric metabolic equivalent value for the exercise.	numeric	--	
is_ai_estimated	Boolean indicating whether the MET value was AI-estimated.	boolean	--	
is_verified	Boolean indicating whether the MET value has been verified.	boolean	--	
confidence	Numeric confidence score for the MET value.	numeric	--	
created_at	Timestamp recording when the record was created in the MET Exercises table.	time stamp	--	
verified_by	Foreign key referencing the profile that verified the exercise value.	uuid	--	FK
verified_at	Timestamp when the exercise value was verified.	time stamp	--	

The MET Exercises table is a reference catalog of exercises with their metabolic equivalent (MET) value, verification state, confidence, and verifier.

Table 21: Chat Rooms

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Chat Rooms table.	uuid	--	PK
participant_one	Primary key component and foreign key referencing the first participant's profile.	uuid	--	FK
participant_two	Primary key component and foreign key referencing the second participant's profile.	uuid	--	FK
created_at	Timestamp recording when the record was created in the Chat Rooms table.	time stamp	--	

The Chat Rooms table represents a one-to-one conversation between two profiles, identified by the two participants and creation timestamp.

Table 22: Chat Messages

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Chat Messages table.	uuid	--	PK
room_id	Foreign key referencing the chat room containing the message.	uuid	--	FK
sender_id	Foreign key referencing the profile that sent the message.	uuid	--	FK
content	Text body of the chat message.	text	--	
created_at	Timestamp recording when the record was created in the Chat Messages table.	time stamp	--	

The Chat Messages table stores individual messages within a chat room, including sender, content, and creation timestamp.

Table 23: Notifications

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Notifications table.	uuid	--	PK
user_id	Foreign key referencing the profile receiving the notification.	uuid	--	FK
title	Short text title of the notification.	text	--	
body	Text body content of the notification.	text	--	
read	Boolean indicating whether the notification has been read.	boolean	--	
created_at	Timestamp recording when the record was created in the Notifications table.	time stamp	--	

The Notifications table stores push or in-app notifications delivered to a user, including title, body, read state, and creation timestamp.

Table 24: Predictions

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Predictions table.	uuid	--	PK
member_id	Foreign key referencing the member the prediction concerns.	uuid	--	FK
metric_name	Text name of the predicted fitness metric.	text	--	
predicted_value	Decimal value predicted by the analytics model.	decimal	10,2	
predicted_date	Date the prediction is made for.	date	--	
confidence	Decimal confidence score of the prediction, between 0 and 1.	decimal	4,3	
created_at	Timestamp recording when the record was created in the Predictions table.	time stamp	--	

The Predictions table stores predictive analytics outputs for a member, including the metric, predicted value, prediction date, and model confidence.

Table 25: Admin Logs

Field Name	Description	Field Type	Field Size	Type of Key
id	Primary key uniquely identifying each row in the Admin Logs table.	uuid	--	PK
admin_id	Foreign key referencing the administrator who performed the action.	uuid	--	FK
action	Text describing the administrative action taken.	text	--	
target_type	Text categorizing the type of target entity.	text	--	
target_id	UUID identifier of the target entity.	uuid	--	
details	JSON object containing structured details about the action.	json	--	
created_at	Timestamp recording when the record was created in the Admin Logs table.	time stamp	--	

The Admin Logs table records administrative actions performed in the system, including the acting admin, action type, target, and structured details.

3.10 GANTT CHART

Gantt chart

Activities	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov
Planning												
Data Gathering (Observation & Face-to-Face Interview)												
Meeting												
Identification of appropriate software to use												
UI/UX Design & Prototyping												
Database & System Architecture												
Admin Website Development												
Member Mobile App Development												
Trainer Mobile App Development												
System Testing												
Implementing Changes												
Implementing Feedback												
Consultation												
Finishing Documentation												

Figure 29: Gantt Chart (Work Plan)

Figure 29 presents the Gantt chart of the work plan for the development of the proposed Intelligent Fitness Progress Monitoring Application of Triple J Fitness Center. The chart covers a twelve-month development period running from December to November, and it lists in its activities column the fourteen major activities undertaken by the proponents. Each shaded cell marks the month or months during which an activity was carried out, so the chart shows the sequence and the overlapping of the work rather than only its start and end dates.

The project began with Planning in December to January, followed by Data Gathering in January to February, where the proponents conducted direct observation of the existing manual tracking and administrative practices of the center and face-to-face interviews with the management, the trainers, and the members to establish the functional and non-functional

requirements of the proposed system. Group meetings were held from December to June to sustain the coordination of the proponents throughout the requirements and design stages. The appropriate software to use was identified in March, after which the design activities were undertaken: the UI/UX Design and Prototyping of the application and the Database and System Architecture of the proposed system.

Development was then carried out in three concurrent workstreams corresponding to the three roles of the system. The Admin Website Development covered the administrative modules such as enrollment confirmation, member management, and reporting; the Member Mobile App Development covered workout logging, QR check-in and check-out, meal logging with AI food analysis, and progress monitoring; and the Trainer Mobile App Development covered the trainer dashboard, assigned member monitoring, and feedback and rating. These workstreams ran alongside one another so that the shared database and the interfaces of the three platforms remained consistent.

System Testing followed the completion of each workstream, together with the Implementing Changes activity for the defects and adjustments identified during testing. The Implementing Feedback activity then covered the revisions requested by the users and the panel during beta testing, and Consultation was conducted to review the progress of the study with the adviser and the panel. The work plan concludes with Finishing Documentation, in which the proponents completed the manuscript, the figures, the data dictionary, and the other supporting documents of the study.

3.11 ROLES AND RESPONSIBILITTIES

Member	Designation	Core Responsibilities
 John Paul L. Hoquiton	Project Manager Full - Stack Documentation	Manages the overall development of the system; coordinates tasks and monitors project progress; develops and integrates both front-end and back-end components; manages system functions and database integration; and assists in preparing and organizing project documentation.
 Kasandra A. Calma	Documentation (main) Quality Assurance System Tester Asistant Front - End	Prepares and organizes the project documentation; assists in designing the user interface and user experience; supports front-end development; conducts system testing and quality assurance; identifies and reports system issues; and assists in ensuring that the system meets the required specifications.

3.12 EVALUATION PROCEDURE

The quality model serves as a framework for evaluating the proposed software system. It provides criteria for measuring the quality, effectiveness, and overall performance of the system based on the requirements of its intended users and stakeholders. The evaluation helps determine whether the system performs its intended functions and provides a satisfactory experience to its users.

The proposed Intelligent Fitness Monitoring Application Using Predictive Analytics will be evaluated using a questionnaire adapted from the ISO/IEC 25010 Software Quality Model. The evaluation will determine the level of acceptability of the system in terms of functional suitability, performance efficiency, usability, reliability, security, maintainability, and portability. The respondents will include selected users and stakeholders of Triple J Fitness Center, who will be given an opportunity to use and assess the system based on their experience.

The evaluation will focus on the identified software quality characteristics to determine whether the system meets its intended requirements. A four-point Likert scale will be used to measure the responses of the evaluators, ranging from Strongly Disagree (1) to Strongly Agree (4). The responses collected will then be summarized and analyzed to determine the overall level of acceptability of the proposed system.

Table 26 ISO 25010 Guideline

Evaluation Criteria	Description
Functional Sustainability	Measures whether the system provides the required functions for managing members, trainers, fitness records, goals, workout logs, and fitness progress predictions
Performance Efficiency	Measures how well the system performs its functions in terms of response time, processing speed, and use of system resources.
Compatibility	Measures the system's ability to operate properly with the required devices, browsers, software, and other system components without affecting its functions.
Reliability	Measures the system's ability to perform consistently and correctly while minimizing errors, failures, and interruptions during use.
Usability	Measures how easy the system is to understand, learn, navigate, and use for administrators, trainers, and members.
Maintainability	Measures how easily the system can be maintained, corrected, updated, and modified when changes or errors occur.
Evaluation Design	Describes the method used to evaluate the system, including the selected respondents, evaluation questionnaire, software quality criteria, and rating scale used to gather and analyze feedback.

The proposed system will be evaluated based on selected software quality criteria to determine its overall quality, effectiveness, and acceptability. The evaluation will focus on Functional Suitability, Performance Efficiency, Compatibility, Reliability, Usability, Maintainability, and Evaluation Design to assess whether the system meets the needs and requirements of its intended users.

RESPONDENTS OF THE STUDY

The respondents of the study will consist of selected administrators, trainers, and members of Triple J Fitness Center who will participate in the evaluation of the proposed Intelligent Fitness Monitoring Application Using Predictive Analytics. The respondents will assess the system based on its functional suitability, performance efficiency, compatibility, reliability, usability, and maintainability. Their feedback will be used to determine whether the system meets the requirements and needs of its intended users.

EVALUATION INSTRUMENT

To gather the necessary data, the researchers will utilize a structured Evaluation Rating Scale delivered via Google Forms. The instrument is adapted from the ISO/IEC 25010 quality standards, with questions translated into a user-friendly format for both technical and non-technical respondents.

RATING SCALE

The survey will utilize a 5-point Likert Scale to measure the level of agreement for each criterion:

Table 27 Likert Scale

Scale	Range	Interpretation
5	4.21 – 5.00	Highly Acceptable
4	3.41 – 4.20	Acceptable
3	2.61 – 3.40	Neutral
2	1.81 – 2.60	Unacceptable
1	1.00 – 1.80	Totally Unacceptable

DATA ANALYSIS METHOD

The data collected will be analyzed using the Weighted Means Method. This Method will be used to evaluate responses of the respondents gathered from the questionnaires. This will determine the overall level of systems acceptability.

The formula for the weighted means is:

$$WM = \frac{\sum (f \times x)}{N \times f}$$

Where:

- f = frequency of response
- x = Weighted Mean
- ∑(f · x)= Sum of the products of the frequency and the weight
- WM – Weighted Mean

This method allows for a more accurate interpretation of data by considering the significance of each response.

3.13 ETHICAL STANDARDS AND CONSIDERATION

This study upholds ethical research practices to protect the rights, privacy, safety, and interests of the members, trainers, administrators, and other individuals involved in the development of the Intelligent Fitness Monitoring Application Using Predictive Analytics for Triple J Fitness Center.

Before collecting information and conducting system testing, the researchers will secure proper permission and informed consent from the concerned participants. The purpose of the study, the type of information to be collected, and their participation in system testing will be clearly explained, and participation will be voluntary.

Personal information collected for the system, such as member names, ages, fitness goals, body measurements, workout records, food and exercise plans, trainer feedback, and fitness progress data, will be treated with confidentiality. The researchers will follow the Data Privacy Act of 2012 (Republic Act No. 10173), which requires personal information to be processed fairly and lawfully and protected through appropriate safeguards.

Access to member and trainer information will be limited according to their assigned roles in the system. The information and fitness predictions generated by the application will only be used for the purposes of the study and system operations and will not be disclosed to unauthorized individuals or used for unrelated purposes.

The researchers will also ensure that the data, system results, testing results, and documentation are presented accurately and honestly without fabrication, falsification, or misrepresentation. Proper acknowledgment will be given to the sources, references, datasets, and related studies used in the development of the system to maintain academic integrity and avoid plagiarism.

CHAPTER IV

THE PROBLEM AND ITS BACKGROUND

This chapter presents the results and discussion of the developed Intelligent Fitness Monitoring Application Using Predictive Analytics for Triple J Fitness Center. The chapter presents the testing conducted to verify the functionality, performance, reliability, and usability of the system. The findings are discussed based on the specific objectives established for the development of the proposed system.

4.1 DEVELOPMENT AND TESING

UNIT AND INTEGRATION TESTING

Table 28 Unit and Integration Testing for Web Application

Test Case ID	Component Module	Test Condition	Expected Behavior	Actual Result	Status
TC001	Authentication	Admin enters valid login credentials.	The system allows the admin to log in and access the dashboard.	The system successfully logged in the admin and displayed the dashboard.	Passed
TC002	Member Management	Admin adds, edits, or views member information.	The system saves and displays the member information correctly.	The system successfully saved and displayed the member information.	Passed
TC003	Trainer Management	Admin adds, edits, or views	The system saves and	The system successfully	Passed

		trainer information.	displays the trainer information correctly.	saved and displayed the trainer information.	
TC004	Membership Management	Admin adds or updates membership information.	The system saves and displays the correct membership details and status.	The system successfully saved and displayed the membership information.	Passed
TC005	QR and Attendance	Admin scans a member QR code for attendance.	The system identifies the member and records the attendance.	The system successfully identified the member and recorded the attendance.	Passed
TC006	Fitness Goals	Admin or trainer views or updates a member's fitness goals.	The system saves and displays the member's fitness goals correctly.	The system successfully saved and displayed the fitness goals.	Passed
TC007	BMI	User enters the member's height and weight.	The system calculates and displays the member's BMI correctly.	The system successfully calculated and displayed the BMI.	Passed
TC008	Workout Logs	Member workout	The system saves and	The system successfully	Passed

		information is recorded.	displays the workout records correctly.	saved and displayed the workout records.	
TC009	Food and Exercise Plan	Trainer creates a seven-day food and exercise plan.	The system saves the plan and displays it to the assigned member.	The system successfully saved and displayed the plan.	Passed
TC010	Trainer Feedback	Trainer provides feedback for a member.	The system saves and displays the feedback to the intended member.	The system successfully saved and displayed the trainer feedback.	Passed
TC011	Predictive Analytics	System processes available member fitness data.	The system generates the corresponding prediction or recommendation.	The system generates the corresponding prediction or recommendation.	Passed
TC012	Notifications	System sends a notification to a user.	The system displays the notification to the intended user.	The system successfully displayed the notification.	Passed
TC013	Dashboard and Reports	Admin accesses dashboard analytics or reports.	The system displays the appropriate statistics,	The system successfully displayed the dashboard	Passed

			records, and reports.	analytics and reports.	

Legend: Passed – The test case met the expected behavior and no significant errors or issues were encountered during testing.

Table 29 Unit and Integration for Member Mobile Application

Test Case ID	Component Module	Test Condition	Expected Behavior	Actual Result	Status
TC001	User Login	User enters valid account credentials.	The application allows the user to log in and displays the Home screen.	The application successfully logged in the user and displayed the Home screen.	
TC002	User Profile	User views and updates profile information.	The application displays and saves the updated profile information correctly.	The application successfully displayed and saved the profile information.	
TC003	Home Dashboard	User opens the Home screen.	The application displays workout activity, active days, check-ins, AI progress forecast, and fitness progress information.	The application successfully displayed the user's fitness information and progress.	

TC004	Workout	User records a workout activity.	The application saves the workout activity and updates the workout information.	The application successfully saved and updated the workout activity.	
TC005	In & Out	User scans the QR code for gym check-in or check-out.	The application processes the QR scan and records the user's attendance.	The application successfully processed the QR scan and recorded the attendance.	
TC006	Food	User views or records food information.	The application displays and saves the selected food and nutrition information correctly.	The application successfully displayed and saved the food information.	
TC007	Chat	User sends a message to the trainer.	The application sends and displays the message in the chat.	The application successfully sent and displayed the message.	
TC008	BMI	User views their BMI information.	The application calculates and displays the user's BMI and	The application successfully calculated and displayed the	

			category correctly.	BMI and category.	
TC009	Fitness Goals	User views or updates fitness goals.	The application displays and saves the user's fitness goals correctly.	The application successfully displayed and saved the fitness goals.	
TC010	Trainer Feedback	User views feedback provided by the trainer.	The application displays the trainer's feedback correctly.	The application successfully displayed the trainer's feedback.	
TC011	AI Progress Forecast	Application processes the user's fitness data.	The application generates and displays the user's fitness progress forecast.	The application successfully processed the fitness data and displayed the progress forecast.	
TC012	Check-in History	User views monthly check-in information.	The application displays the user's check-in records and monthly progress correctly.	The application successfully displayed the check-in information.	
TC013	BMI Progress	User views BMI progress over time.	The application displays the user's BMI records and	The application successfully displayed the	

			growth over time correctly.	BMI progress over time.	
TC014	Notifications	User opens the notification feature.	The application displays available notifications to the user.	The application successfully displayed the notifications.	
TC015	Settings	User accesses the Settings screen and selects an available feature.	The application displays the selected settings feature and performs the requested action correctly.	The application successfully displayed the settings and performed the selected action.	

Legend: Passed – The test case met the expected behavior and no significant errors were encountered

Table 30 Unit and Integration for Trainer Mobile Application

Test Case ID	Component Module	Test Condition	Expected Behavior	Actual Result	Status
TC001	Trainer Login	Trainer enters valid account credentials.	The application allows the trainer to log in and displays the Trainer Dashboard.	The application successfully logged in the trainer and displayed the Trainer Dashboard.	
TC002	Trainer Dashboard	Trainer opens the Dashboard.	The application displays the number of members, chat status, membership information, workout activity, and at-risk members.	The application successfully displayed the trainer dashboard information.	
TC003	Member Management	Trainer opens the Dashboard.	The application displays the trainer's assigned members and their relevant information.	The application successfully displayed the assigned members and their information.	

TC004	Workout Monitoring	Trainer views the weekly workout activity of members.	The application displays the members' workout activity for the selected period.	The application successfully displayed the weekly workout activity.	
TC005	At-Risk Members	Trainer views the At-Risk Members section.	The application displays members identified as being at risk based on available fitness and attendance data.	The application successfully displayed the at-risk member information.	
TC006	In & Out	Trainer scans or uses the QR feature for member check-in and check-out.	The application processes the QR transaction and records the member's attendance.	The application successfully processed the transaction and recorded the attendance.	
TC007	Chat	Trainer sends a message to a member.	The application sends and displays the message in the chat conversation.	The application successfully sent and displayed the message.	
TC008	Create Plan	Trainer creates a food and	The application saves the plan	The application successfully	

		exercise plan for a member.	and makes it available to the assigned member.	saved and assigned the plan to the member.	
TC009	Record	Trainer records information related to a member's fitness progress.	The application saves and displays the recorded information correctly.	The application successfully saved and displayed the recorded information.	
TC010	Trainer Feedback	Trainer provides feedback to a member.	The application saves the feedback and makes it available to the member.	The application successfully saved and displayed the trainer feedback.	
TC011	Trainer Profile	Trainer views the Profile section.	The application displays the trainer's account information and available profile features.	The application successfully displayed the trainer's profile information.	
TC012	Notifications	Trainer receives a notification.	The application displays the notification to the trainer.	The application successfully displayed the notification.	

TC013	Membership Monitoring	Trainer views membership information shown on the Dashboard.	The application displays the relevant membership status and expiration information.	The application successfully displayed the membership information.	
TC014	Trainer Navigation	Trainer navigates between Dashboard, Members, In & Out, Chat, and Profile.	The application loads each selected section correctly without errors.	The application successfully loaded the selected sections.	

Legend: Passed – The test case met the expected behavior and no significant errors were encountered.

EVALUATED BY IT EXPERTTS

Following the ISO/IEC 25010 Software Quality Model, an evaluation was carried out on the Intelligent Fitness Monitoring Application Using Predictive Analytics for Triple J Fitness Center. The study involved five (5) IT expert respondents, selected based on their background in software development and systems administration. The evaluation focused on Functional Suitability, Performance Efficiency, Compatibility, Reliability, Usability, and Maintainability. Each criterion was measured using a 5-point Likert scale, providing insights into the system's functionality, performance, reliability, and ease of use.

Table 31 Software Evaluation of Functional Suitability Criterion by IT Expert

Criterion	Overall Rating	Interpretation
Functional Suitability		
Performance Efficiency		
Compatibility		
Reliability		
Usability		
Maintainability		
Overall System Rating		

Table # presents….

Evaluation by End Users

The end user….

Table 32 Software Evaluation of Functional Suitability Criterion

Criterion	Sub- Criterion	Rating	Interpretation

Table # presents…..

Table 33 Software Evaluation of Performance Efficiency Criterion

Criterion	Sub- Criterion	Rating	Interpretation

Table # presents…..

USER ACCEPTANCE TESTING

Using the ISO/IEC 25010 Software Quality Model, an evaluation was conducted on the Intelligent Fitness Monitoring Application Using Predictive Analytics to assess its software quality, functionality, and overall performance. The study involved [number of respondents] respondents, consisting of selected administrators, trainers, and members of Triple J Fitness Center. Their feedback was essential in determining how well the system

supports fitness monitoring and management and how effectively it meets the needs and expectations of its intended users.

The evaluation focused on Functional Suitability, Performance Efficiency, Compatibility, Reliability, Usability, and Maintainability. Each criterion was measured using a 5-point Likert scale, which provided a basis for determining the level of acceptance of the developed system.

4.2 EVALUATION RESUTS

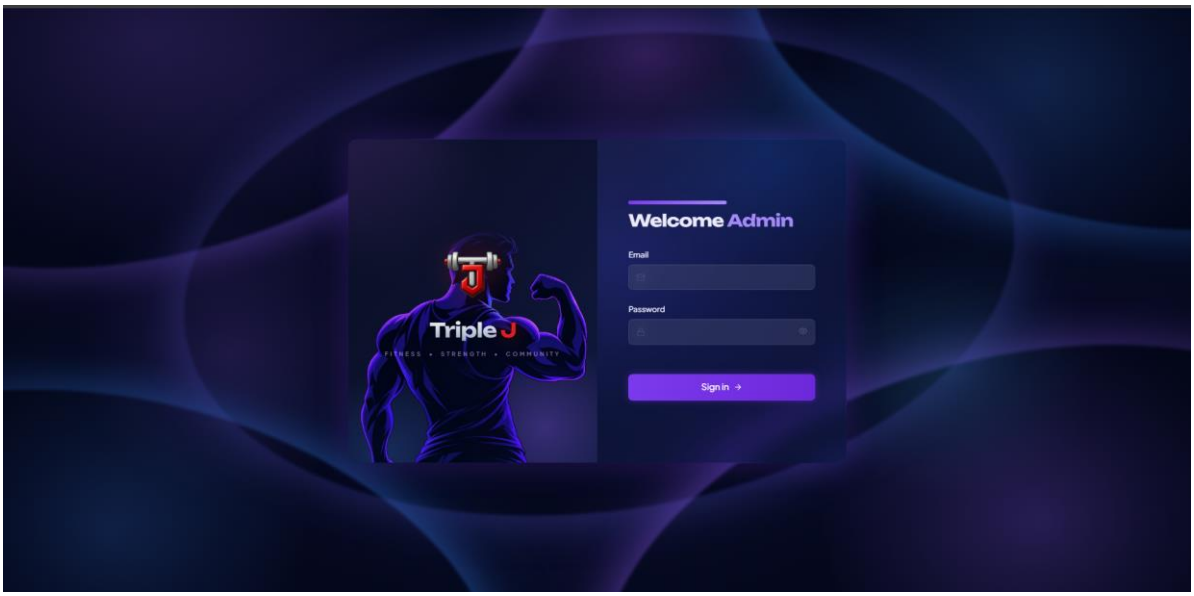
CHAPTER V

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

APPENDICES

User Manual

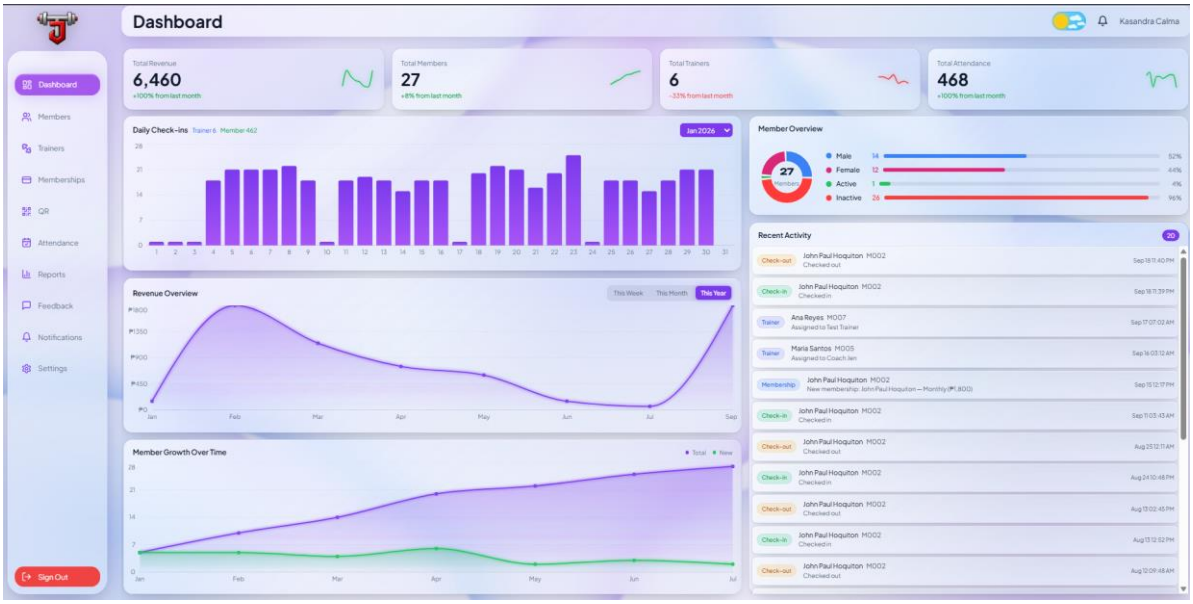
ADMIN



1. Open Admins Website

2. Admin Log in's


Dashboard
Kasandra Calma
Dashboard
Members
Trainers
Memberships
QR
Attendance
Reports
Feedback
Notifications
Settings
Sign Out
Total Revenue
6,460
Total Members
27
Total Trainers
6
Total Attendance
468
Daily Check-ins
Jan 2026
Member Overview
Male
Female
Active
Inactive
Recent Activity
Revenue Overview
This Week
This Month
This Year
Member Growth Over Time
Total
New


Dashboard
Kasandra Calma
Dashboard
Members
Trainers
Memberships
QR
Attendance
Reports
Feedback
Notifications
Settings
Sign Out
Total Revenue
6,460
Total Members
27
Total Trainers
6
Total Attendance
468
Daily Check-ins
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Member Overview
Male
Female
Active
Inactive
Recent Activity
Revenue Overview
This Week
This Month
This Year
Member Growth Over Time
Total
New